

Summary of Changes:

1. Proof of concept #4: redid analysis to use KVL to reach an equation where resistors in parallel have equal voltage, measurement records voltage for both R3 and R3 instead of node 1 voltage and current, compared values of R3 and R4 voltage in table of results instead of Node 1 voltage

2. Proof of concept #1: handwritten analysis allowed for Lab 1

3. Proof of concept #2: handwritten analysis allowed for Lab 1 me to

Question 2
Voltage divider in a series circuit 1.5 / 2 pts

+ 2 pts I can demonstrate how voltage drops across resistors in a series circuit (voltage divider) and prove it with good documentation.

✓ + 1.5 pts I can demonstrate how voltage drops across resistors in a series circuit (voltage divider), but my documentation needs improvement.

+ 0.5 pts Proof of concept well documented, but major technical errors are present.

+ 0 pts Error in technical content (error in mathematical analysis, simulation, experiment, or discussion OR major parts missing)

+ 0 pts Labels missing or inconsistent between schematic and results

+ 0 pts Results not legible or not presented in a logical order

+ 0 pts Completely missing or incorrect.

+ 0 pts Repeated submission. One submission per group!

No handwritten work.

Question 4
Voltage in a parallel circuit 0.5 / 2 pts

+ 2 pts I can demonstrate how voltage is dropped across resistors in a parallel circuit and prove it with good documentation.

+ 1.5 pts I can demonstrate how voltage is dropped across resistors in a parallel circuit, but my documentation needs improvement.

✓ + 0.5 pts Proof of concept well documented, but major technical errors are present.

✓ + 0 pts Error in technical content (error in mathematical analysis, simulation, experiment, or discussion OR major parts missing)

✓ + 0 pts Labels missing or inconsistent between schematic and results

Question 1
KVL, KCL and Ohm's Law 1.5 / 2 pts

+ 2 pts I can demonstrate that KVL, KCL and Ohm's Law hold in a real circuit and prove it with good documentation.

✓ + 1.5 pts I can demonstrate that KVL, KCL and Ohm's Law hold in a real circuit, but my documentation needs improvement.

+ 0.5 pts Proof of concept well documented, but major technical errors are present.

+ 0 pts Error in technical content (error in mathematical analysis, simulation, experiment, or discussion OR major parts missing)

+ 0 pts Labels missing or inconsistent between schematic and results

✓ + 0 pts Results not legible or not presented in a clear or logical order

+ 0 pts Completely missing or incorrect.

+ 0 pts Repeated submission. One submission per group!

No handwritten equations.

ECSE 1010 Lab 1- Omega

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Proof of Concepts- Part A

Evan: (1) Prove Ohm's Law, KCL, and KVL

Jacob: (2) Prove the concept of a voltage divider in a series circuit, (3) Prove the concept of how current flows in a series circuit

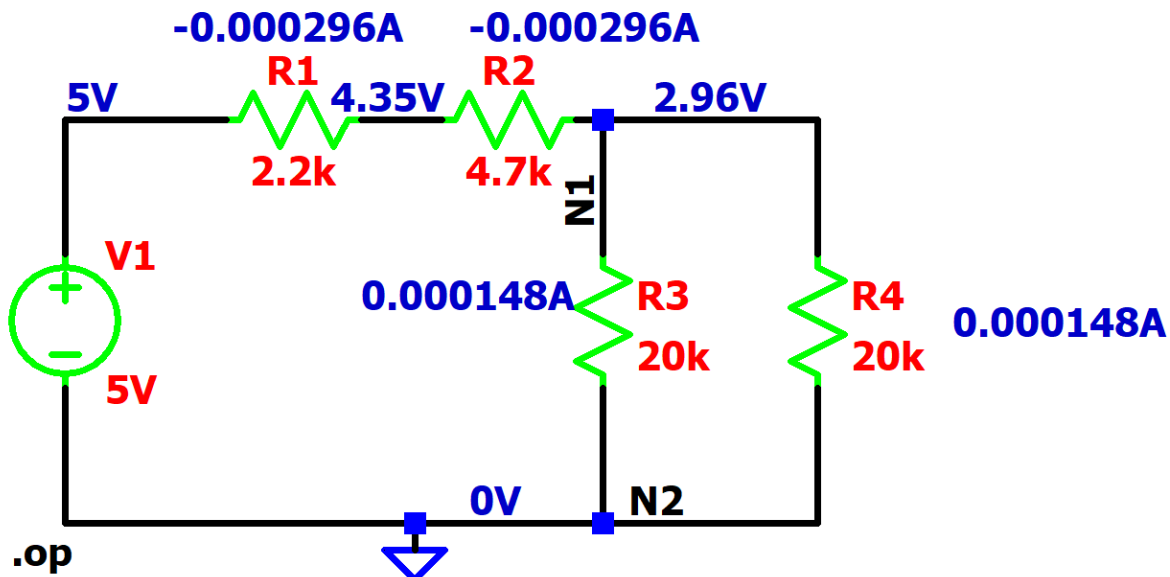
Kimberly: (4) Prove the concept of a voltage across a parallel circuit, (5) Prove the concept of a current divider in a parallel circuit.

1. Prove Ohm's Law, KCL, and KVL

Circuit Schematic:

LTspice circuit schematic of the circuit you will use to prove the concept.

Clearly label all nodes in the you will reference, for example.



Description:

Short description of the concept you will prove and how you will prove it.

Specify which variables in the circuit will be analyzed.

Ohm's Law describes the relationship between the voltage, current, and resistance through a resistor. The equation for this is $V=IR$, where V is the voltage, I is the current, and R is the resistance of the resistor. To prove this, I will build the basic circuit shown above and measure the voltage across the first resistor, R_1 , with a voltmeter, the current through it with an ammeter, and then use the resistance of 2.2 k ohms to prove that the Ohm's Law equation is valid. I will do this for all the remaining resistors in the circuit too: R_2 , R_3 , and R_4 . Kirchhoff's Current Law describes the relationships of current coming into and out of a node. I will prove this by measuring the voltage and current and using the resistance of the resistor to confirm the equation is true. It states the sum of the currents into the node will equal the sum of the currents leaving the node. To prove Kirchhoff's Current Law, KCL, I will use these currents that I measured and, considering the directions of the current, will assure that the currents into Node 1, N_1 , sum up to the sum of the currents leaving Node 1. Kirchhoff's Voltage Law, KVL, states the voltages across all components in a loop must add up to 0. To prove Kirchhoff's Voltage Law, KVL, I will use the 2 loops, L_1 and L_2 , and consider the orientation of the voltage source and the resistors to confirm that the sum of all the voltages across all of the components will equal 0.

Analysis:

$$\begin{aligned} \text{Loop 1: } 5 - V_{R_1} - V_{R_2} - V_{R_3} &= 0 \\ \downarrow \\ V_{R_1} + V_{R_2} + V_{R_3} &= 5 \\ \text{Loop 2: } V_{R_3} - V_{R_4} &= 0 \\ \text{KCL } N_1: I_{R_2} - I_{R_3} - I_{R_4} &= 0 \\ \downarrow \\ \frac{V_{R_2}}{4.7k} - \frac{V_{R_3}}{20k} - \frac{V_{R_4}}{20k} &= 0 \\ \text{KCL } N_2: I_{R_3} + I_{R_4} - I_{R_1} &= 0 \\ \downarrow \\ \frac{V_{R_3}}{20k} + \frac{V_{R_4}}{20k} - \frac{V_{R_1}}{2.2k} &= 0 \end{aligned}$$
$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & \frac{1}{4.7k} & -\frac{1}{20k} & -\frac{1}{20k} \\ -\frac{1}{2.2k} & 0 & \frac{1}{20k} & \frac{1}{20k} \end{bmatrix} \begin{bmatrix} V_{R_1} \\ V_{R_2} \\ V_{R_3} \\ V_{R_4} \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$
$$\begin{bmatrix} V_{R_1} \\ V_{R_2} \\ V_{R_3} \\ V_{R_4} \end{bmatrix} = \begin{bmatrix} 0.6509 \\ 1.3905 \\ 2.956 \\ 2.956 \end{bmatrix}$$

Loop 1: $I_{R_1}(2.2k) + I_{R_2}(4.7k) + I_{R_3}(20k) = 5$

Loop 2: $I_{R_3}(20k) - I_{R_4}(20k) = 0$

Node 1: $I_{R_2} - I_{R_3} - I_{R_4} = 0$

Node 2: $I_{R_3} + I_{R_4} - I_{R_1} = 0$

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$$\begin{bmatrix} 2.2k & 4.7k & 20k & 0 \\ 0 & 0 & 20k & -20k \\ 0 & 1 & -1 & -1 \\ -1 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} I_{R_1} \\ I_{R_2} \\ I_{R_3} \\ I_{R_4} \end{bmatrix} = \begin{bmatrix} 5 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} I_{R_1} \\ I_{R_2} \\ I_{R_3} \\ I_{R_4} \end{bmatrix} = \begin{bmatrix} 2.874 \times 10^{-4} A \\ 2.874 \times 10^{-4} A \\ 1.437 \times 10^{-4} A \\ 1.437 \times 10^{-4} A \end{bmatrix}$$

Ohm's Law

$R_1: 0.6509 V = (2.874 \times 10^{-4} A)(2.2k \Omega)?$

$R_2: 1.3905 V = (2.874 \times 10^{-4} A)(4.7k \Omega)?$

$R_3: 2.956 V = (1.437 \times 10^{-4} A)(20k \Omega)?$

$R_4: 2.956 V = (1.437 \times 10^{-4} A)(20k \Omega)?$

Equation and short description.

Clearly describe how you are applying the concept to your circuit. You should calculate specific values to compare to simulations and experiment.

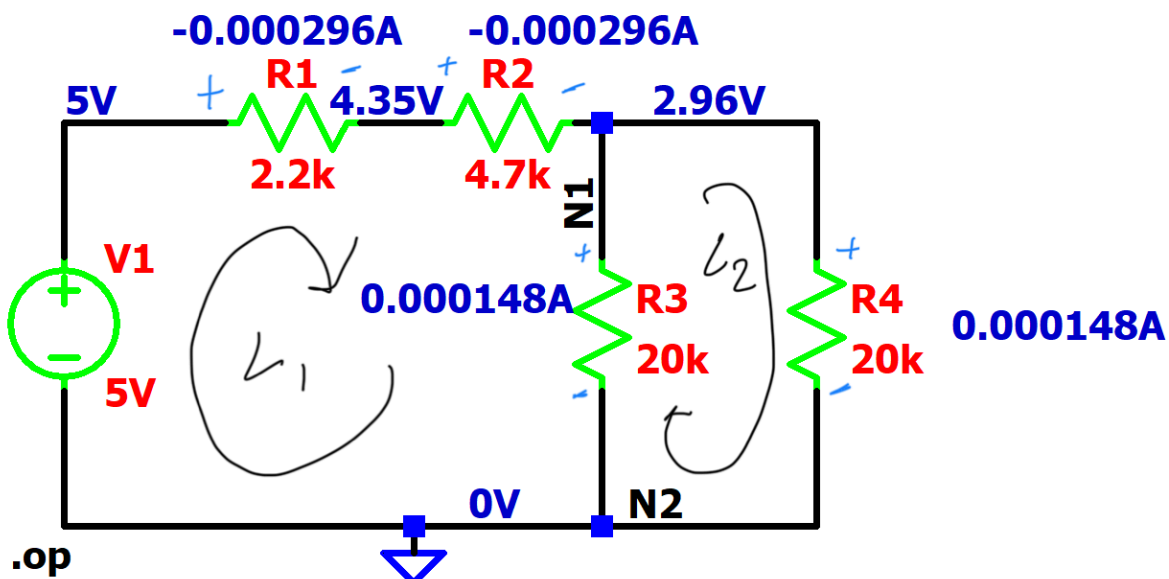
The equation of Ohm's Law is $V=IR$, where V is the voltage across a resistor, I is the current, and R is the resistor. Unfortunately, Ohm's Law is needed to be able to perform KVL and KCL to solve for the voltages and currents in the resistors. Above, I used KVL and KCL to solve for both the current and the voltage across each resistor and then checked the consistency of these values with Ohm's Law.

Simulation:

Screenshot of simulation and short description.

Clearly labeled simulation results with nodes and/or input/output that matches with schematic above. Any important portions of output are identified (i.e. the point at which a comparator switches is circles and/or point to with labeled arrow for easy identification).

Shown in the simulation below are all of the resistors in which I will be measuring the voltages and currents of in order to prove Ohm's, Kirchoff's Current, and Kirchoff's Voltage Laws. The voltages, currents, and resistance values across all components are clearly labeled in the schematic along with the nodes I will be referencing. Also, the polarities for the resistors are displayed.



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* C:\Users\ewwit\Documents\LTspice\Draft10.asc

--- Operating Point ---

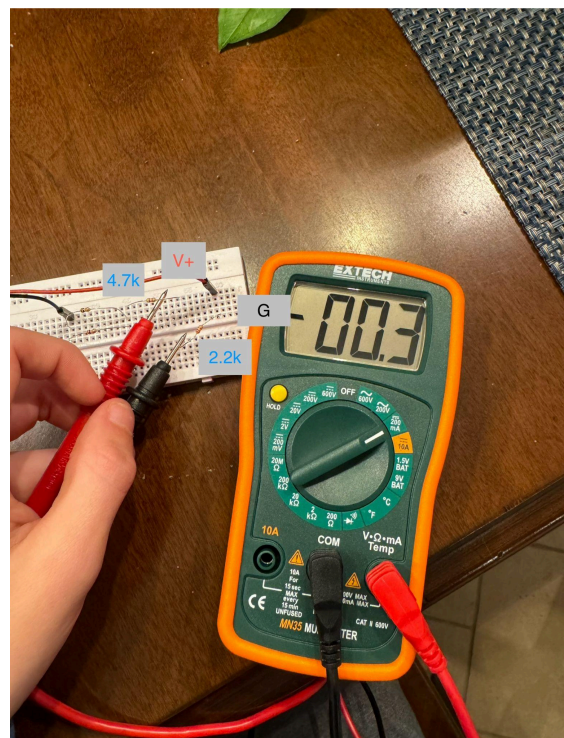
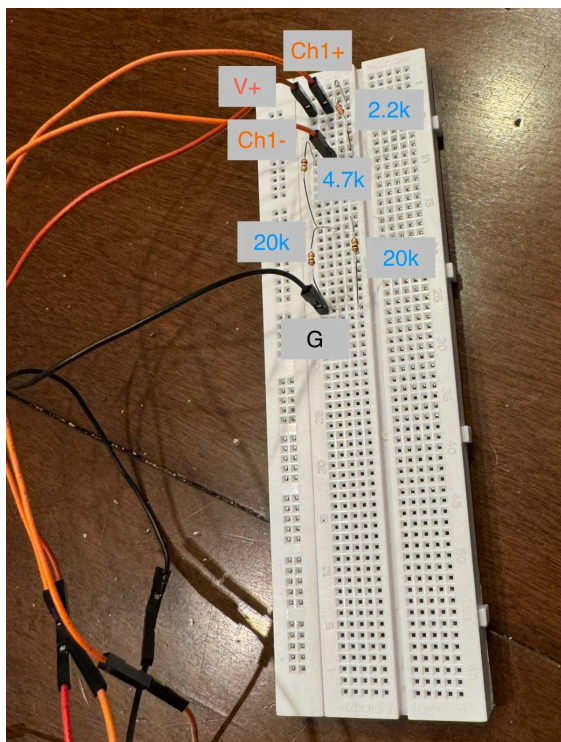
V(n001):      5          voltage
V(n002):      4.34911    voltage
V(n1):        2.95858    voltage
I(R1):        -0.000295858 device_current
I(R2):        -0.000295858 device_current
I(R3):        0.000147929 device_current
I(R4):        0.000147929 device_current
I(V1):        -0.000295858 device_current
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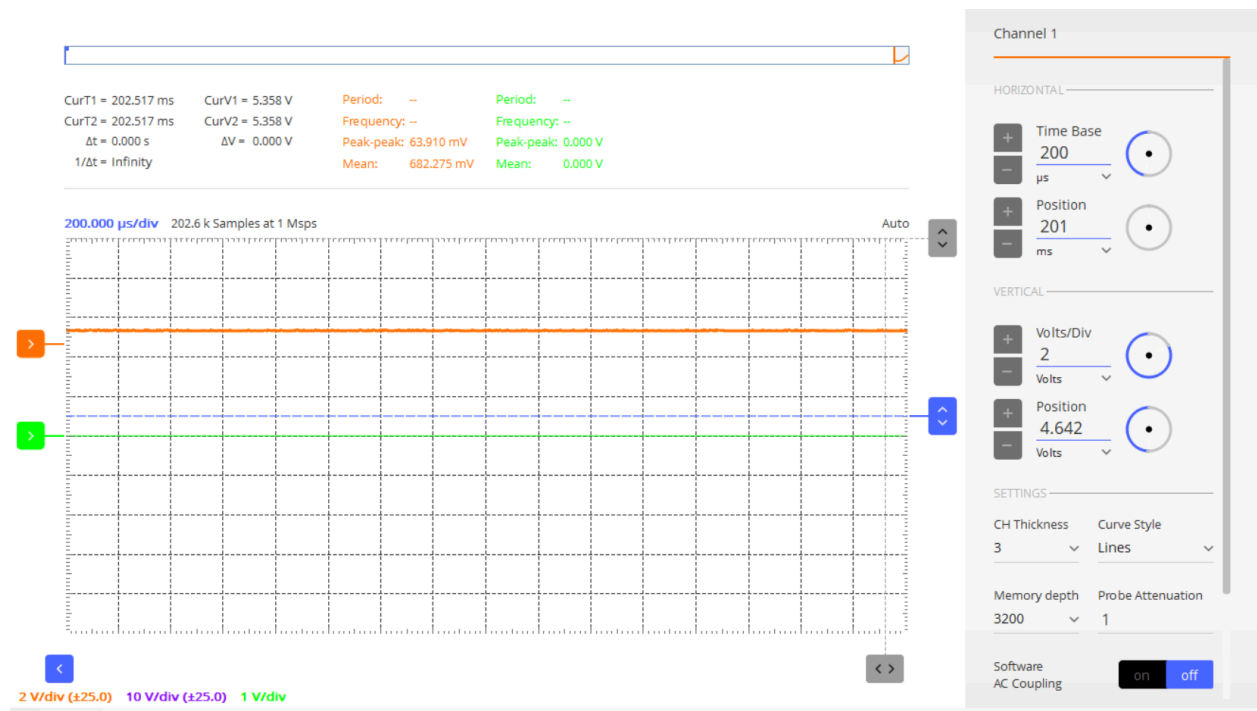
Measurement:

Screenshot of Waveforms output from circuit above and short description.

Remember to clearly show all axes in a measurement plot. Also identify any important portions of the output.

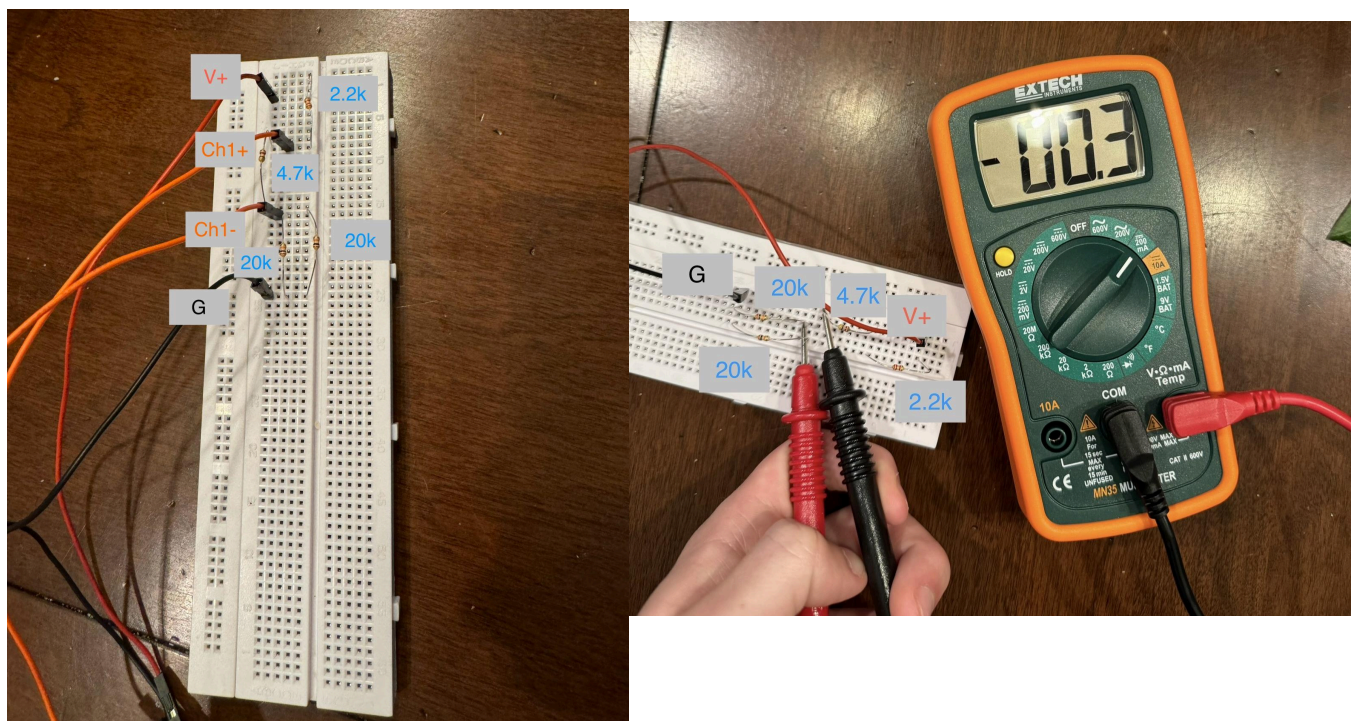
R1

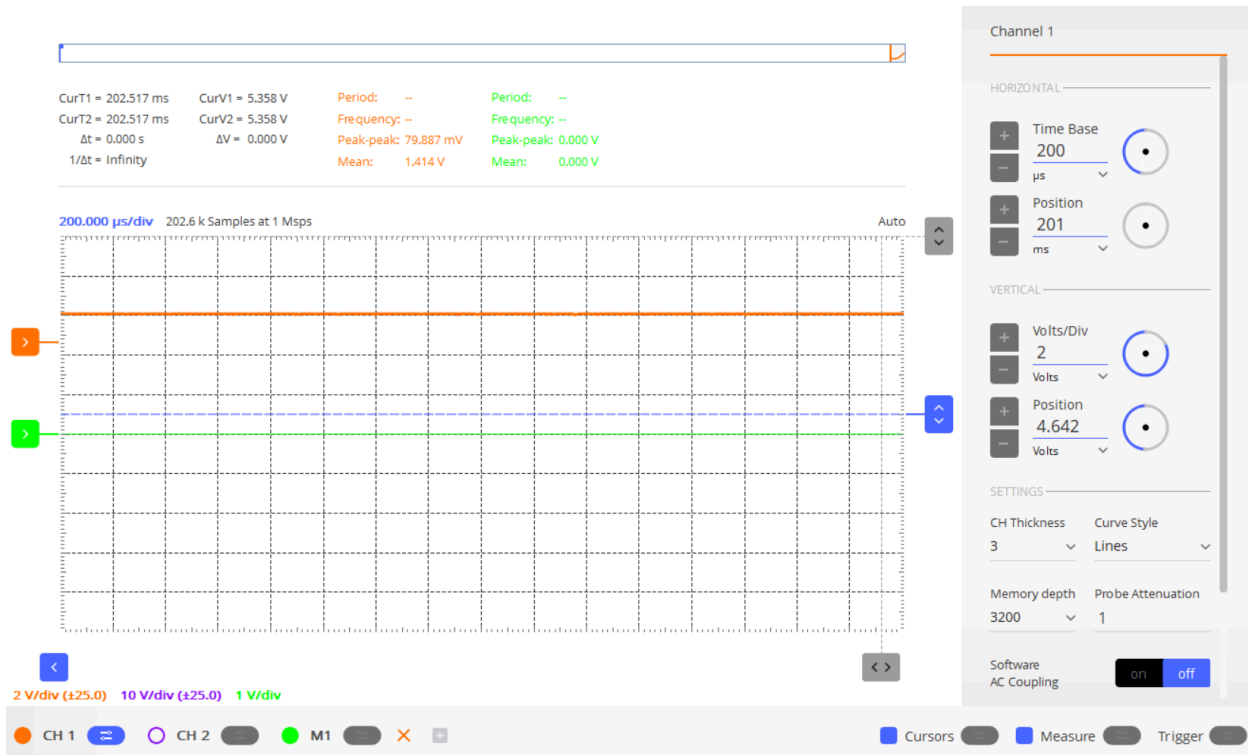




Above is the circuit schematic along with the current and voltage measured through the 2.2k resistor, R1. The current is measured as -0.3 mA and the voltage as 0.682 V.

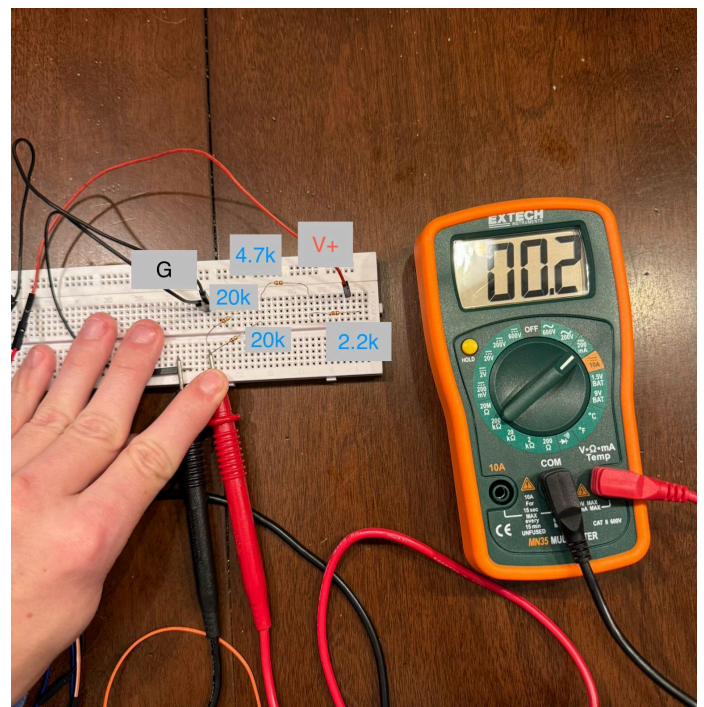
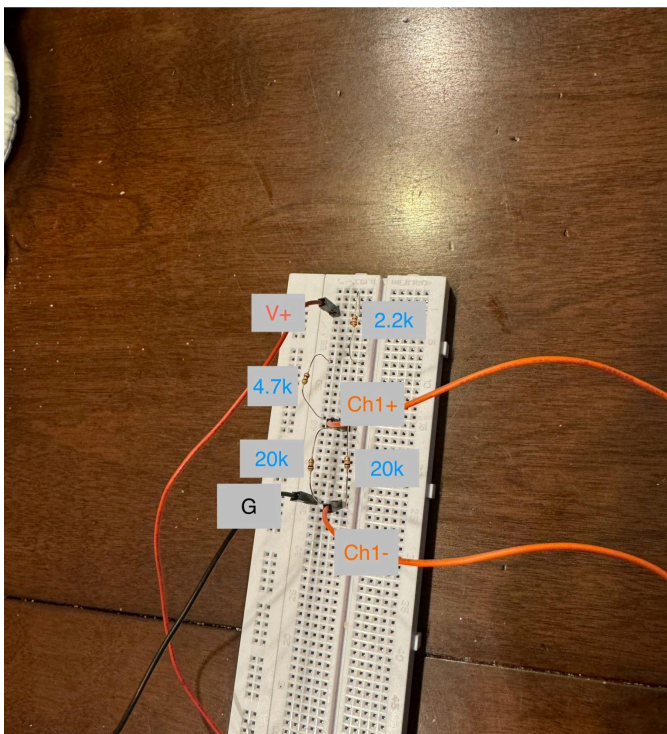
R2

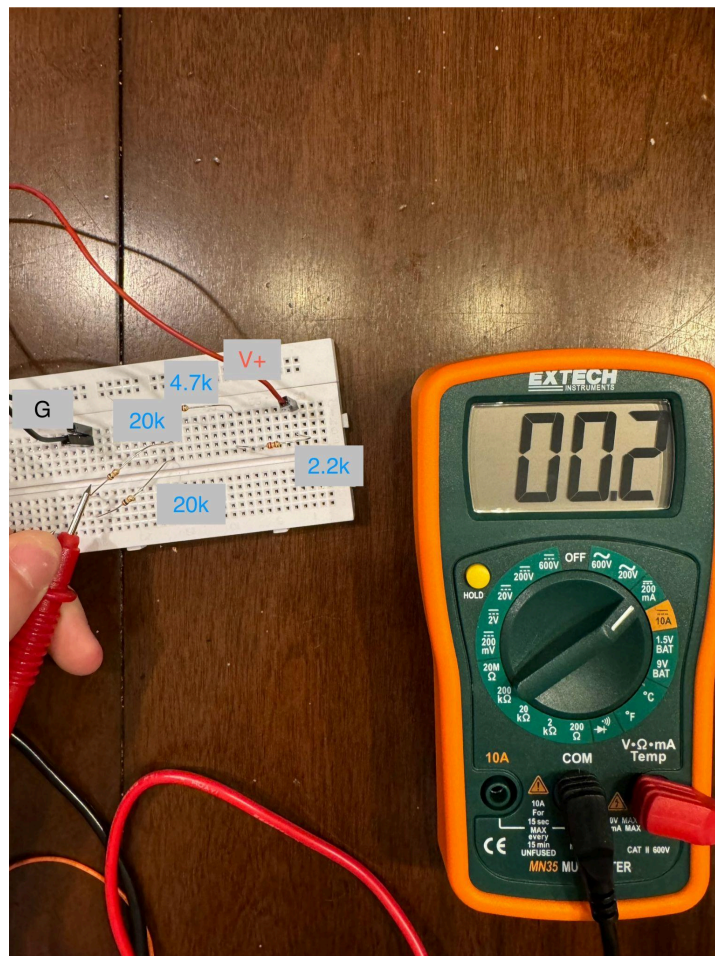




Above is the circuit schematic along with the current and voltage measured through the 4.7k resistor, R2. The current is measured as -0.3 mA and the voltage as 1.414 V.

R3 and R4.





Above is the circuit schematic along with the current and voltage measured through the 20k resistors, R3 and R4. The current measured through both resistors is 0.2 mA and the voltage is 2.985V.

Discussion (and answer related questions in Alpha Lab):

In the analysis section, we used a combination of KVL, KCL, and Ohm's Law to solve for the voltages and currents through each resistor. Using the simulated values, Ohm's Law is consistent as for R_1 , $5-4.35=(0.000296)(2200)$, for R_2 , $4.35-2.96=(0.000296)(4700)$, and for R_3 and R_4 , $2.96=(0.000148)(20000)$; since the left and right sides are all equal, this proves Ohm's law as the left side represents the voltage and the right side represent the current multiplied by the resistance. Using the simulated values, this will come out to $0.296\text{mA}=0.148\text{mA}+0.148\text{mA}$ which is consistent and proves KCL. This could also be done for Node 2, N_2 , even if it is connected to ground. Using the simulated values, the equation turns into $5-(5-4.35)-(4.35-2.96)-2.96=0$ and then simplified to $5-0.65-1.39-2.96=0$ which is consistent with $0=0$ and proves KVL for loop 1. For loop 2, the equation will look like $V_{R3}-V_{R4}=0$ and with the simulated values, $2.96-2.96=0$ which is consistent with KVL as $0=0$. Using the experimental values, Ohm's law is proven as for R_1 , $0.682\text{V}=(0.3\text{mA})(2.2\text{k})$, for R_2 , $1.414\text{V}=(0.3\text{mA})(4.7\text{k})$, and for R_3 and R_4 , $2.985\text{V}=(0.2\text{mA})(20\text{k})$. Obviously these values aren't completely accurate for the experimental values, but since the resistances and the general components have some degree of error, as long as the equations are fairly close, usually within about 3%, then the concepts are proven.

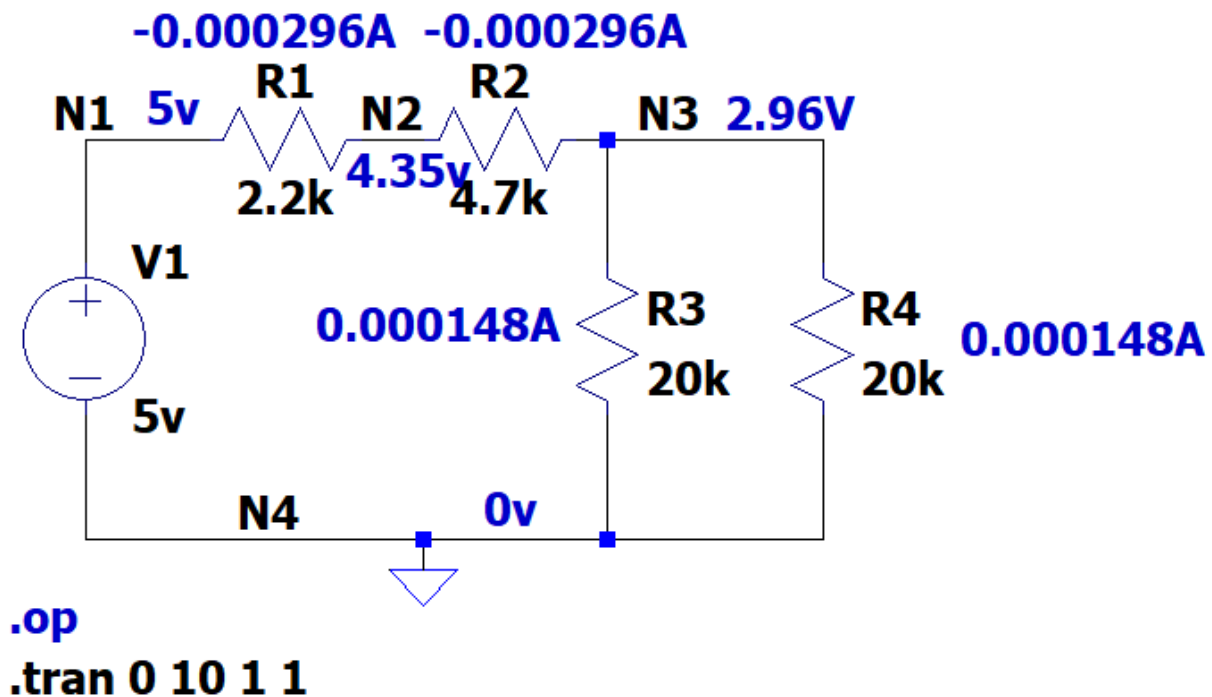
	Analysis	Simulation	Experiment	% Err
VR1 (V)	0.6509	0.65	0.682	4.78%
IR1 (mA)	0.2874	0.296	0.3	4.38%
VR2 (V)	1.3905	1.39	1.414	1.69%
IR2 (mA)	0.2874	0.296	0.3	4.38%
VR3 (V)	2.956	2.96	2.985	0.98%
IR3 (mA)	0.1437	0.148	0.2	39.18%
VR4 (V)	2.956	2.96	2.985	0.98%
IR4 (mA)	0.1437	0.148	0.2	39.18%

2. Prove the concept of a voltage divider in a series circuit

Circuit Schematic:

LTspice circuit schematic of the circuit you will use to prove the concept.

Clearly label all nodes in the you will reference, for example.



Description:

Short description of the concept you will prove and how you will prove it.

Specify which variables in the circuit will be analyzed.

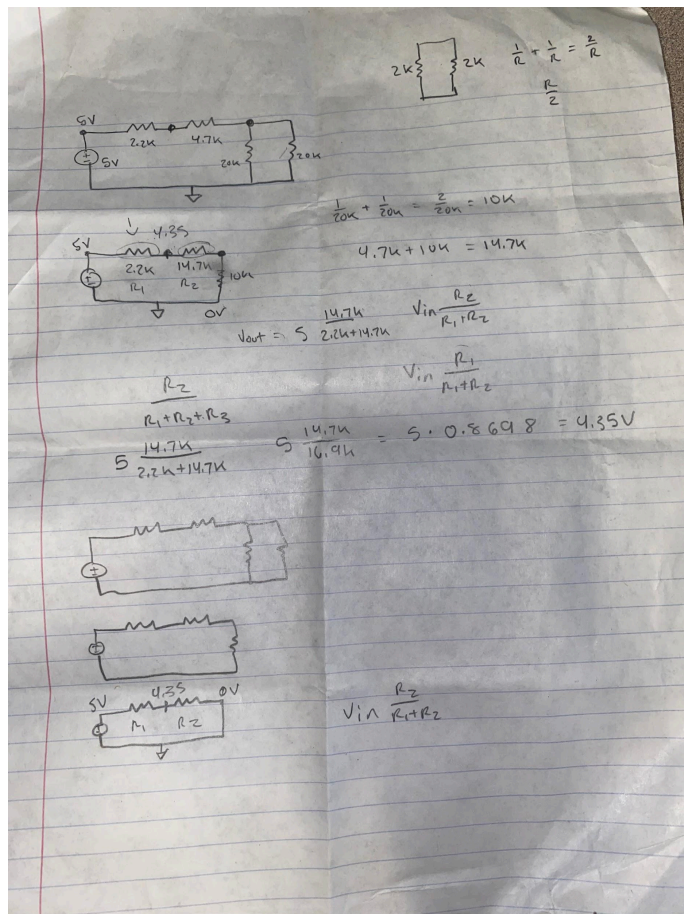
This concept is the distribution of voltage through components of a divider, in this case two resistors in a series, with the input voltage distributed over both resulting in the voltage output. The equation for this concept is that of $V_{out} = V_{in} * R2 / (R1 + R2)$. In this equation V_{out} is

voltage out, V_{in} being voltage in, R_2 being the resistance of resistor 2, and finally R_1 being the resistance of resistor 1. To prove this concept I will create the model with my breadboard and Analog Discovery 3 utilizing the circuit schematic above and show that through waveforms, LTSpice, and the equation above how voltage is divided across a resistor.

Analysis:

Equation and short description.

Clearly describe how you are applying the concept to your circuit. You should calculate specific values to compare to simulations and experiment.

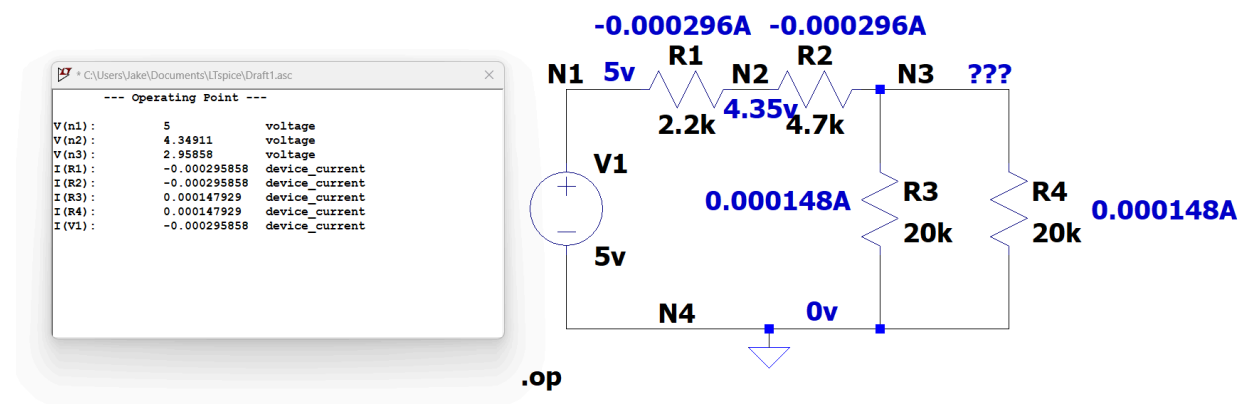
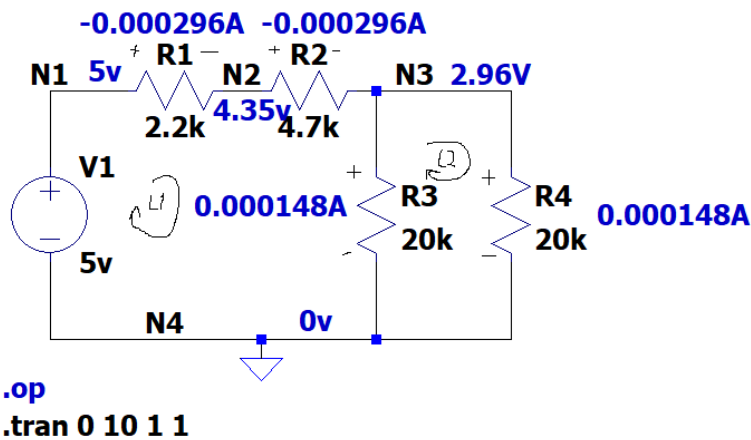


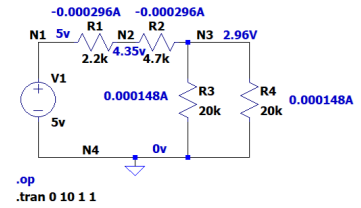
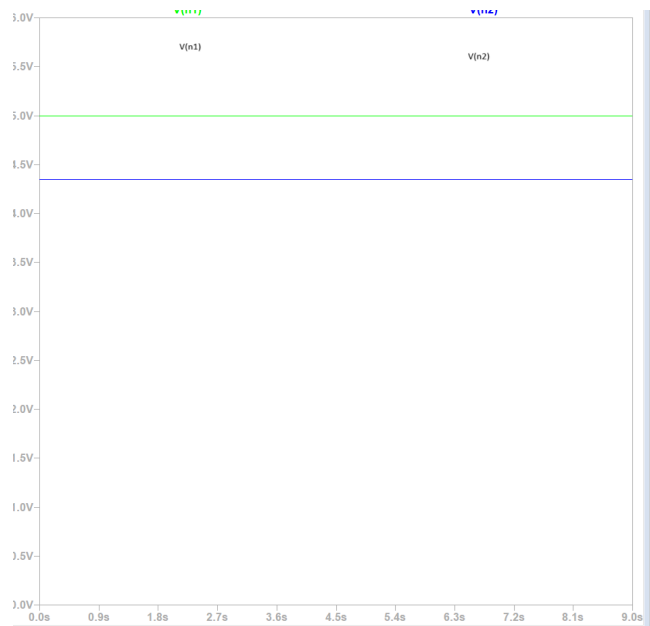
Simulation:

Screenshot of simulation and short description.

Clearly labeled simulation results with nodes and/or input/output that matches with schematic above. Any important portions of output are identified (i.e. the point at which a comparator switches is circles and/or point to with labeled arrow for easy identific

As Demonstrated with the simulations, the voltage divided across R1 and R2 is clearly shown and found through the usage of the equation $V_{out} = V_{in} R_2 / (R_1 + R_2)$ and the .op feauture in LTspice I will find the Voltage of N3 which turns out to be 2.96 and how the voltage is divided across resistor 2 with the knowledge of node 2 having a voltage of 4.35.

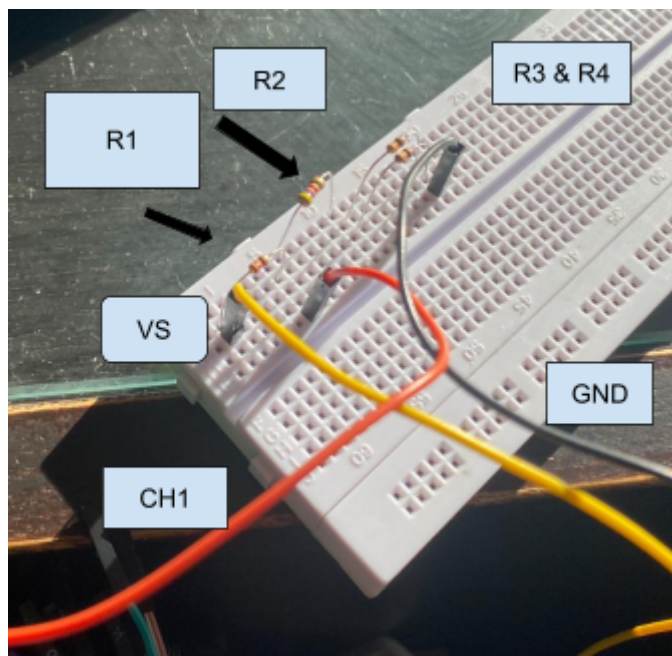


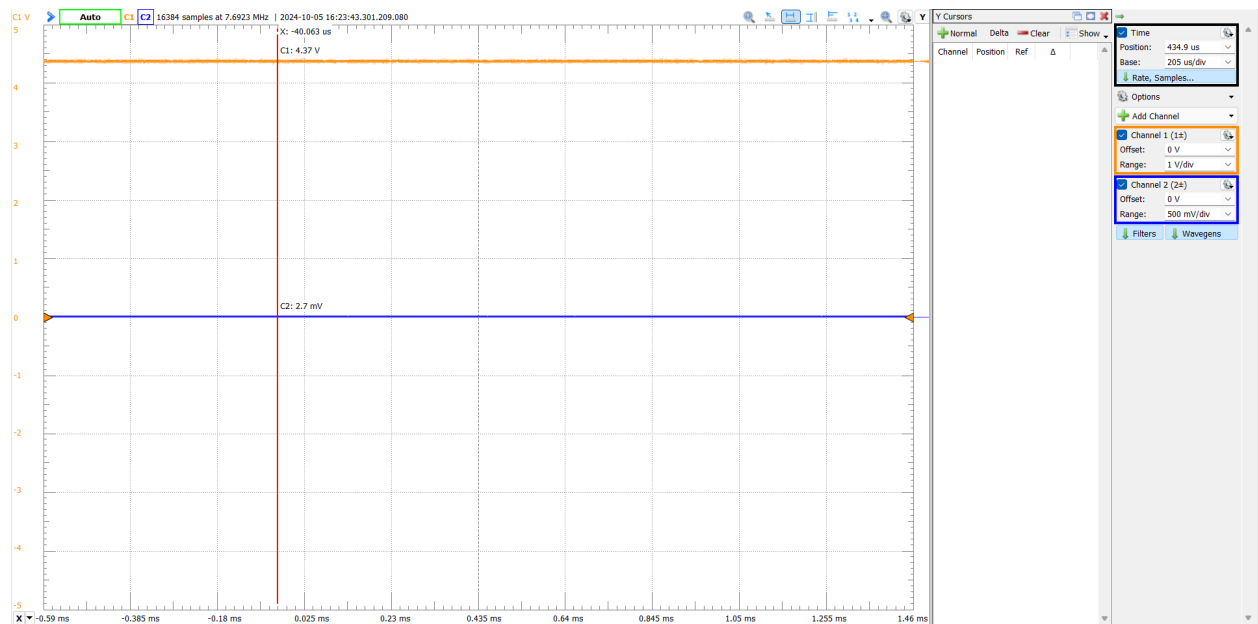


Measurement:

Screenshot of Waveforms output from circuit above and short description.

Remember to clearly show all axes in a measurement plot. Also identify any important portions of the output





Using Waveforms Oscilloscope with a Voltage source of 5 we see that for N2 reads 4.37V roughly equivalent to 4.35V as predicted on LT Spice.

Discussion (and answer related questions in Alpha Lab):

Comparison of Analysis, Simulation and Measurement results. Both a simple summary of results (like a numerical chart of values) and a simple description that details if the results are as you expect. Also include any speculation as to why they may be different from one another if they are different. What variation is too much for example...explore this.

Include a table that compares mathematical, simulation, and experimental results.

	Analysis	Simulation	Experiment	% Err
Voltage of N2	4.35	4.35	4.37	0.46%

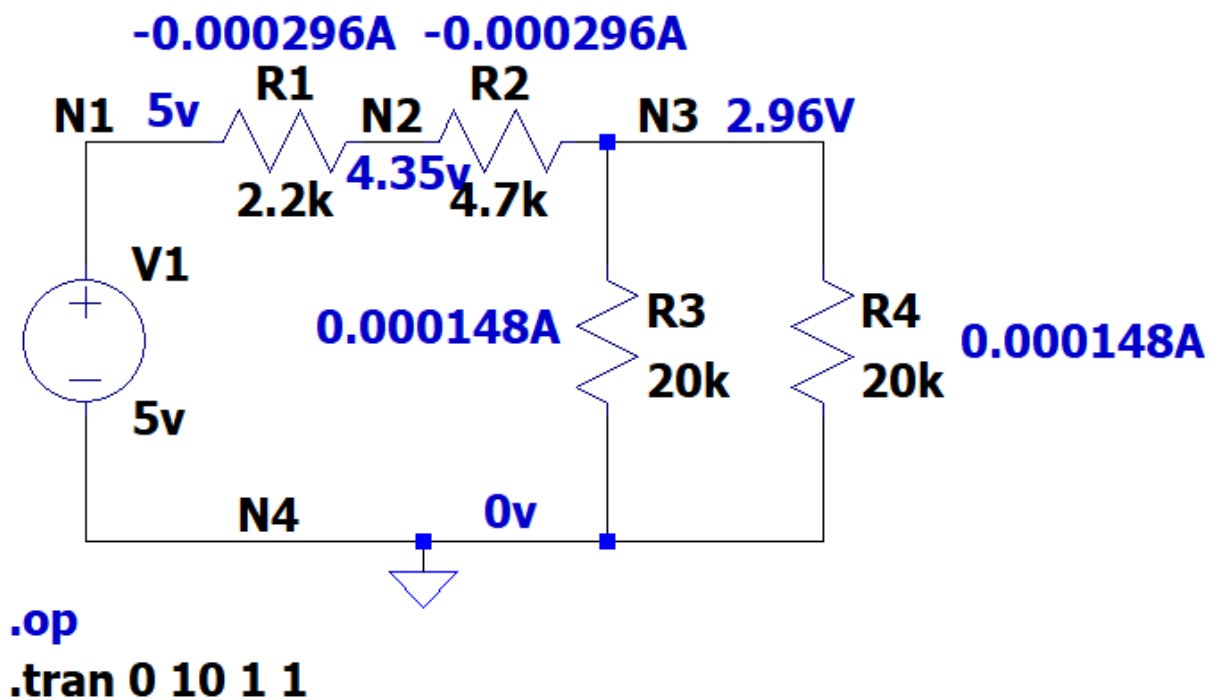
The results illustrate the drop of voltage across a resistor with trying to find Node 3's voltage, already knowing the drop across Resistor 1 and Node 2's voltage. The results gathered are as expected, with both the equation done by hand, the oscilloscope measurements, and LTspice's table of values all matching up. There are some slight differences, but this can generally be attributed to just variation in input when done by hand and fluctuation in current.

3. Prove the concept of how current flows in a series circuit

Circuit Schematic:

LTspice circuit schematic of the circuit you will use to prove the concept.

Clearly label all nodes in the you will reference, for example.



Description:

Short description of the concept you will prove and how you will prove it.

Specify which variables in the circuit will be analyzed.

With series circuits, components are connected from end to end, voltage coming in from the power source and divided among the components of the circuit based on their resistances, and based on KVL we know that the amount of those resistors added up equal to the voltage, while

the current remains constant. The equation for this concept is $I = V/R_{total}$ with I being the current, V being the voltage in, and R_{total} being the sum of all individual resistances found in the series. To prove this concept I will use the circuit above and find the currents through different points using LTSpice while testing a physical model using a voltmeter to find the voltage drop across resistors. Then using Ohm's law and the equation $V=IR$ I will calculate the expected voltage drops and compare.

Analysis:

Equation and short description.

Clearly describe how you are applying the concept to your circuit. You should calculate specific values to compare to simulations and experiment.

The equation is $I = V/R_{total}$, I being the current, V being the voltage in, and R_{total} being the sum of all individual resistances found in the series. To find the current we take the voltage in, in this case 5 volts, and divide that by the total resistance in the circuit, in this circuit that adding up to 16.9K ohms.

Equation

$$I = 5 / R_1 + R_2 + R_3,4$$

$$I = 5 / 2.2K + 4.7K + 10K$$

$$I = 5 / 16.9K$$

$$I = 0.29585mA$$

$$I = 0.000296A$$

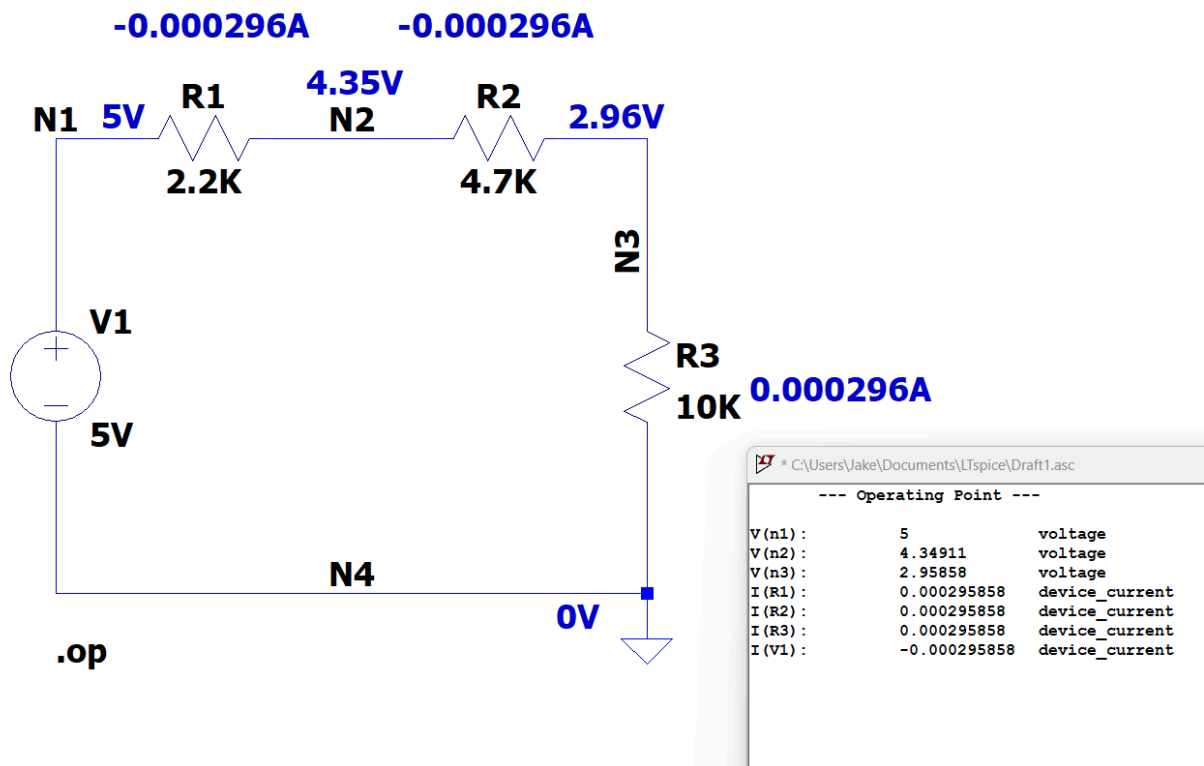
Due to nature of current $I = 0.000296A$ or $I = 0.296A$

Simulation:

Screenshot of simulation and short description.

Clearly labeled simulation results with nodes and/or input/output that matches with schematic above. Any important portions of output are identified (i.e. the point at which a comparator switches is circles and/or point to with labeled arrow for easy identification).

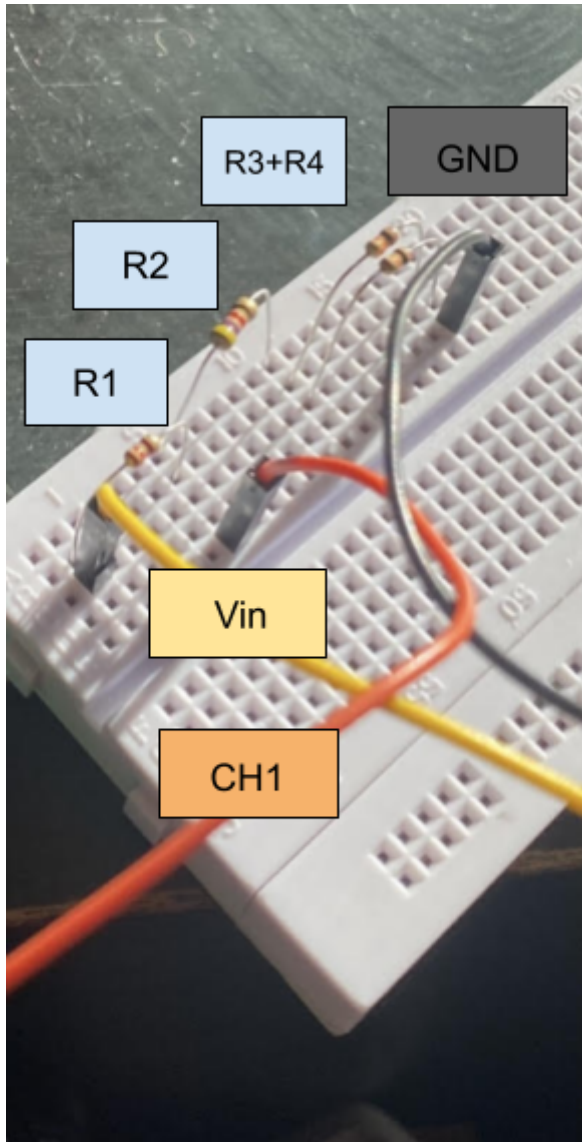
Here is a simulation of the circuit on LTSpice which shows the current between R1 and R2, that being I(R1) on the graph which shows how the current remains the same through the circuit. The only reason I(V1) is different is due to the polarity of where the current is entering and exiting.



Measurement:

Screenshot of Waveforms output from circuit above and short description.

Remember to clearly show all axes in a measurement plot. Also identify any important portions of the output.



Made the circuit above on a breadboard like this:

Where V_{in} is the voltage in which is 5V,
Ch1 is measuring current between R1 and R2

$R1 = 2.2k\Omega$

$R2 = 4.7k\Omega$

$R3 = 20k\Omega$

$R4 = 20k\Omega$

And GND is ground.



Using transient analysis feature on LTspice shows the current between R1 and R2 and confirms the math used in analysis as well as the data from the simulation.

Discussion (and answer related questions in Alpha Lab):

Comparison of Analysis, Simulation and Measurement results. Both a simple summary of results (like a numerical chart of values) and a simple description that details if the results are as you expect. Also include any speculation as to why they may be different from one another if they are different. What variation is too much for example...explore this.

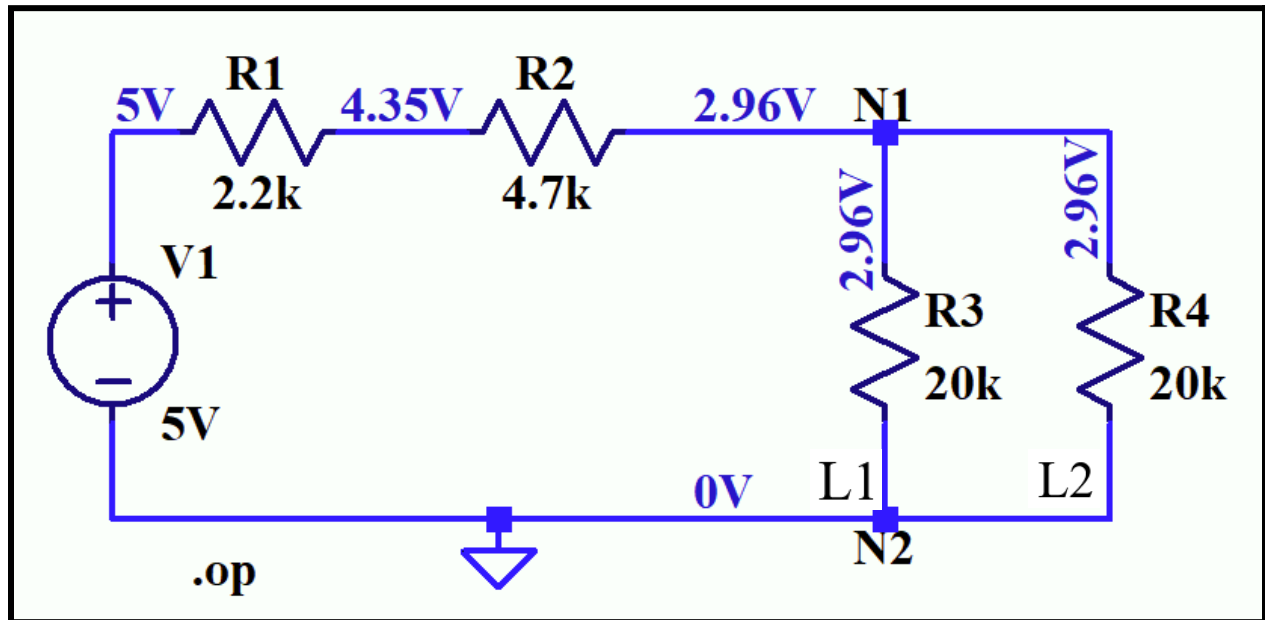
Include a table that compares mathematical, simulation, and experimental results.

	Analysis	Simulation	Measurement	Percent Error
Current through R1	$I = 0.000296\text{A}$	$I = 0.000296\text{A}$	$I = 0.00029586\text{A}$	0.047%
Current through R2	$I = 0.000296\text{A}$	$I = 0.000296\text{A}$	$I = 0.00029586\text{A}$	0.047%
Current through R3	$I = 0.000296\text{A}$	$I = 0.000296\text{A}$	$I = 0.00029586\text{A}$	0.047%

From the table above we can see that the simulation and analysis very very closely correlate with the experimental data gathered and that current, throughout the circuit remains the same. The error present is most likely just fluctuation in voltage in which effects the current flowing through the circuit

4. Prove the concept of voltage across a parallel circuit

Circuit Schematic:



Description:

From this circuit, I will prove how the voltage of R_3 and R_4 , which are resistors in parallel, are equal using Kirchhoff's Voltage Law where the sum of the voltages in a closed loop must equal to zero. In this circuit, the resistors in parallel are in loop 2 (L2) and so the sum of V_3 and V_4 should equal to zero, and therefore, $V_3 = V_4$.

Analysis:

To prove how voltage in a parallel circuit is equal, use KVL.

Apply KVL on loop 2 (L2) so that we know the sum of the voltages of R3 and R4 should equal to 0.

The equation for loop 2 should be:

$$V_3 - V_4 = 0$$

Meaning that $V_3 = V_4$

To find the voltage of either R_3 or R_4 , begin by finding the current at N1. We also know that R_3 and R_4 have the same resistance and equal voltage so they should also have the same current.

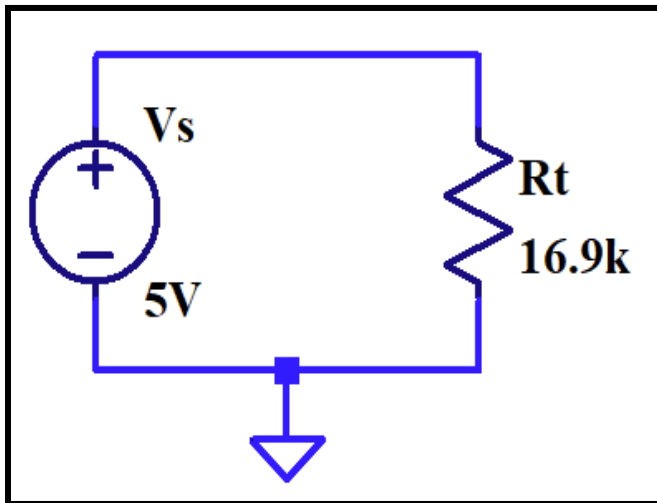
Combining resistors in parallel (R_3 and R_4):

$$R_p = \frac{1}{\frac{1}{R_3} + \frac{1}{R_4}} = \frac{1}{\frac{1}{20k} + \frac{1}{20k}} = 10k\Omega$$

And combining the resistors in series (R_3 and R_4):

$$R_t = 6.9k + 10k = 16.9k\Omega$$

The simplified circuit appears like so:



And by applying Ohm's law where $V = IR$,

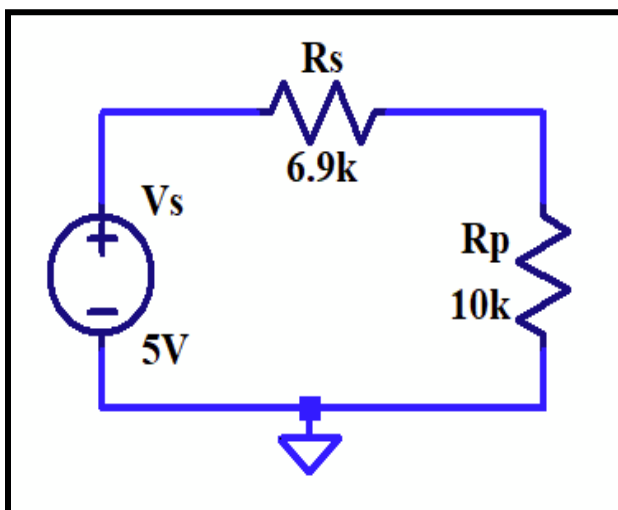
$$V_s = I_t R_t$$

$$5 = I_t (16.9k)$$

$$I_t = 5/16900 = 0.296mA$$

Where I_t is the current of the single resistor simplified circuit, and also, the current through R_1 and R_2 in the original circuit since resistors in series have the same current.

Referring back to the original circuit diagram, we want to now solve for the voltage of N1 (node1) since the voltage at N1 should equal the voltage of both R_3 and R_4 .



Using Ohm's Law again where $V = IR$ to find V_{Rs}

$$V_{Rs} = (I_t)(R_s)$$

$$V_{Rs} = (0.296)(6900)$$

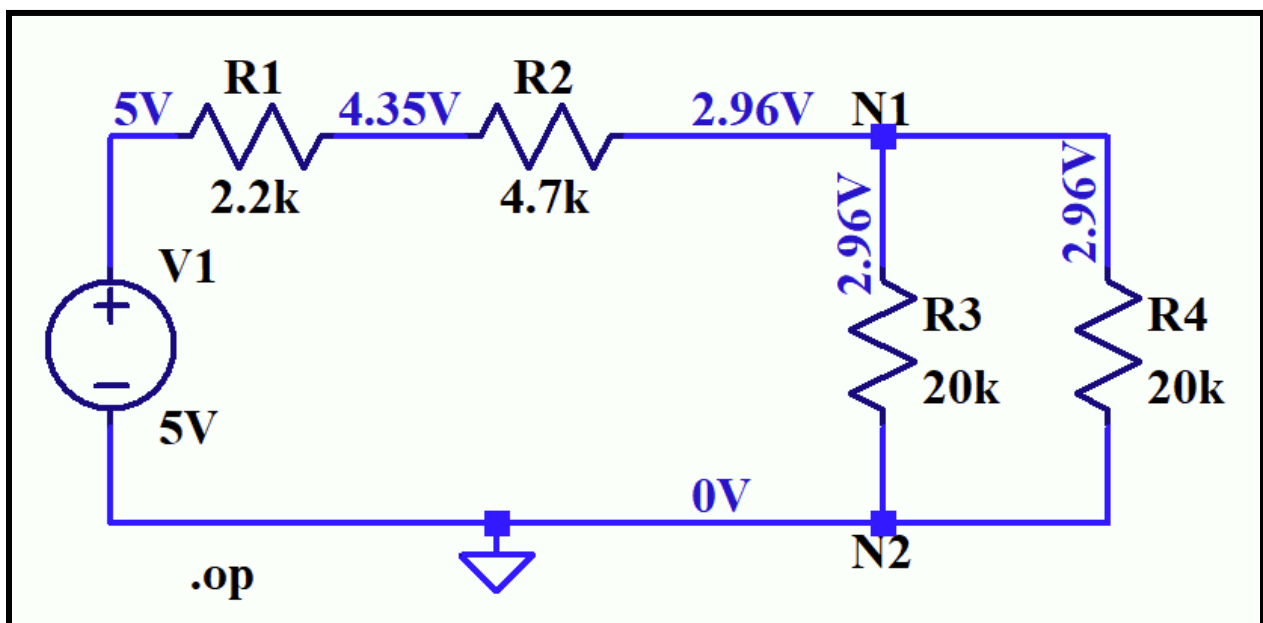
$$V_{Rs} = 2.042V$$

And so, $V_{Rp} = V_s - V_{Rs} = 5 - 2.042 = 2.958V$

Where V_{Rp} is voltage at N1 and so we can declare that the voltages of both R_3 and R_4 which from KVL: $V_3 = V_4$ so $V_3 = V_4 = 2.958V$.

Simulation:

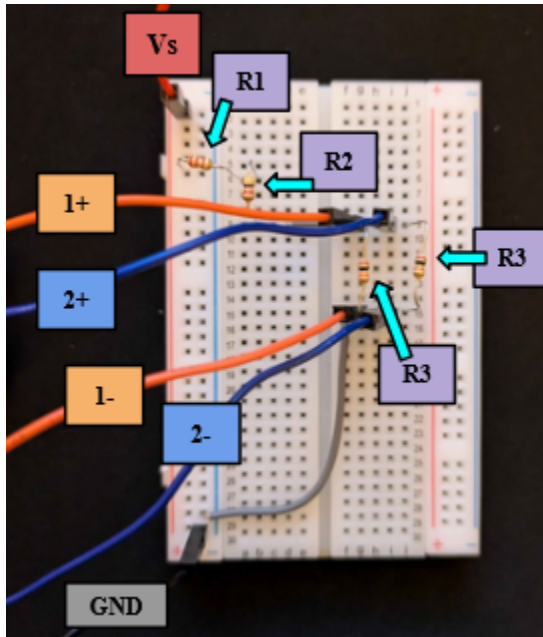
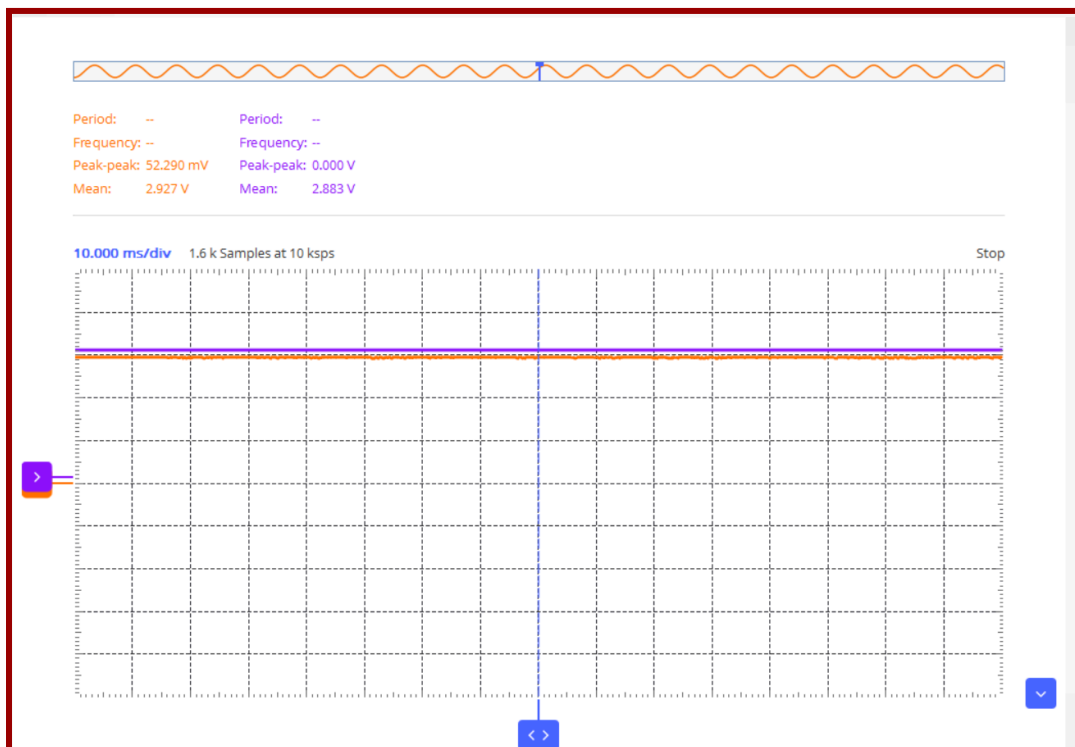
Here is a simulation of the circuit on LTSpice which shows us the voltage at N1 = 2.96V, and the voltages at R3 and R4 are also 2.96V which matches the values we got above from mathematically solving the circuit.



```
--- Operating Point ---
V(n001) :      5      voltage
V(n002) :      4.34911 voltage
V(n1) :        2.95858 voltage
I(R1) :      -0.000295858 device_current
I(R2) :      -0.000295858 device_current
I(R3) :        0.000147929 device_current
I(R4) :        0.000147929 device_current
I(V1) :      -0.000295858 device_current
```

Measurement:

Make the breadboard like this:


$$R1 = 2.2k\Omega$$
$$R2 = 4.7k\Omega$$
$$R3 = 20k\Omega$$
$$R4 = 20k\Omega$$


Measuring using the oscilloscope on Scopy with a 5V power supply shows that the voltage at R3 (Channel 1) is 2.927V and voltage at R4 (Channel 2) is 2.883V.

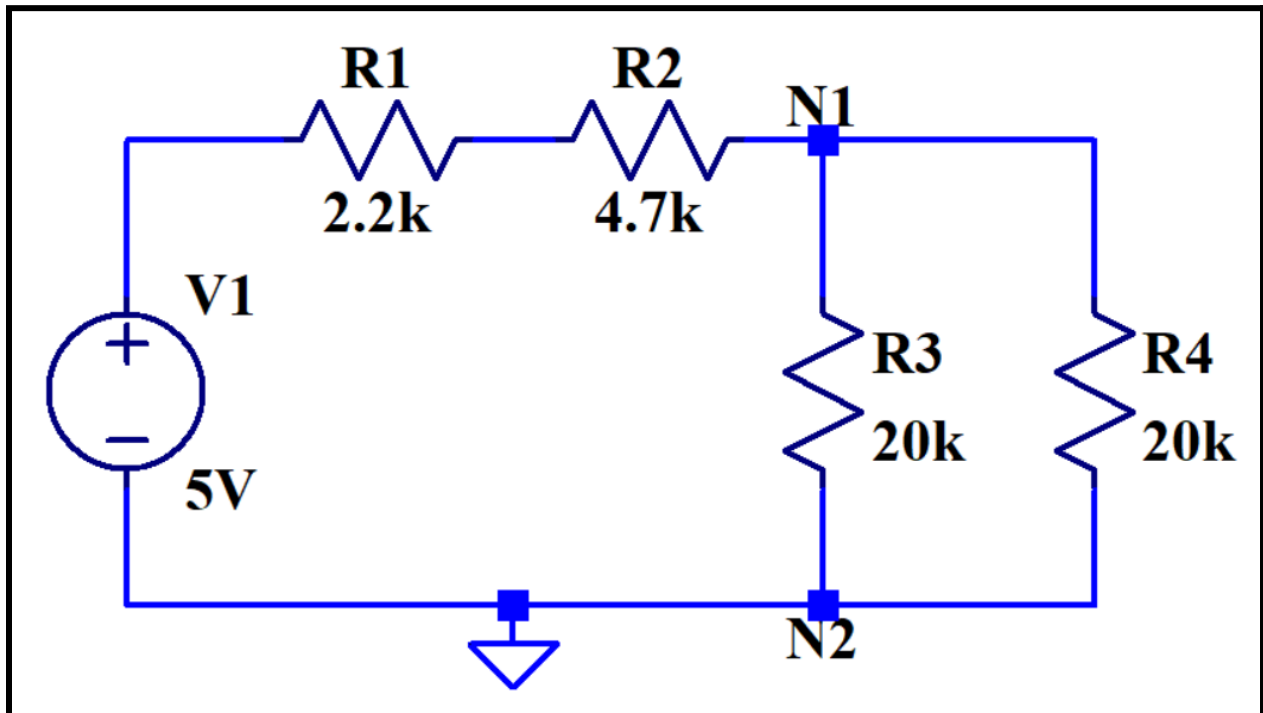
Discussion (and answer related questions in Alpha Lab):

	Analysis	Simulation	Measurement/ Experimental	Percent Error (%)
Voltage of R3 (V)	2.958	2.95858	2.927	1.07%
Voltage of R4 (V)	2.958	2.95858	2.883	2.55%

From the table above, we can see that the values we get from measuring and analysis/simulation are very similar with the greatest percent error being 2.55% for the voltage of R4. With this we can conclude that the values come out to be as we expected them to be and therefore this measurement reinforces that voltage is equal for resistors in parallel.

5. Prove the concept of a current divider across a parallel circuit

Circuit Schematic:



Description:

To prove the concept of a current divider across the circuit above, I will use Ohm's Law to find the current at N1 which will be I_t and use the given values of R3 and R4 to solve for the current of both I_{R3} and I_{R4} using the Current Divider Rule.

Analysis:

First to find I_t , the current at N1, reduce the circuit so that we only see the resistors in parallel (R_3 and R_4). Begin by combining the resistors R_1 and R_2 .

$$R_s = R_1 + R_2 = 2.2k + 4.7k = 6.9k$$

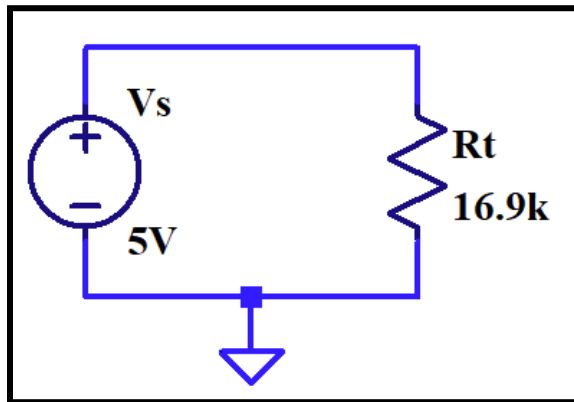
And combine R_3 and R_4 in parallel to be,

$$R_p = \frac{1}{\frac{1}{R_3} + \frac{1}{R_4}} = \frac{1}{\frac{1}{20k} + \frac{1}{20k}} = 10k\Omega$$

And so the total resistance is:

$$R_t = R_s + R_p = 6.9k + 10k = 16.9k\Omega$$

The simplified circuit appears like so:



And by applying Ohm's law where $V = IR$,

$$V_s = I_t R_t$$

$$5 = I_t(16.9k)$$

$$I_t = 5/16900 = 0.296mA$$

Now that we have I_t , we can use the Current Divider Rule:

$$I_n = I_t \frac{R_t}{R_n + R_t}, \text{ where } n \text{ is some value corresponding to the resistor (gets notated as } I_{R3} \text{ or } R_3)$$

To find current of R_3 in a circuit with 2 resistors in parallel, substitute in so that

$$I_{R3} = (0.296)\left(\frac{20k}{20k+20k}\right) = 0.148mA$$

And similarly for R_4 ,

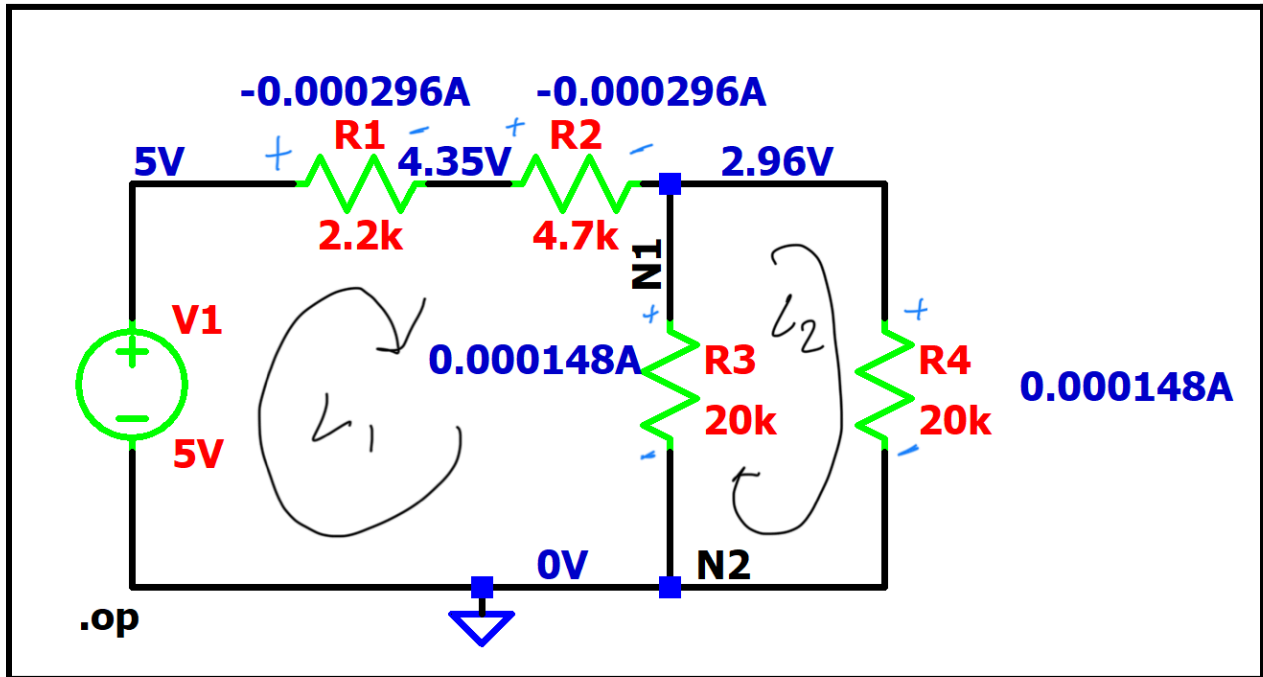
$$I_{R4} = (0.296)\left(\frac{20k}{20k+20k}\right) = 0.148mA$$

And we can verify this with Kirchoff's Current Law where at N1, the current I_t should equal the sum of the currents I_{R3} and I_{R4} .

$$I_t = I_{R3} + I_{R4} = 0.148 + 0.148 = 0.296mA$$

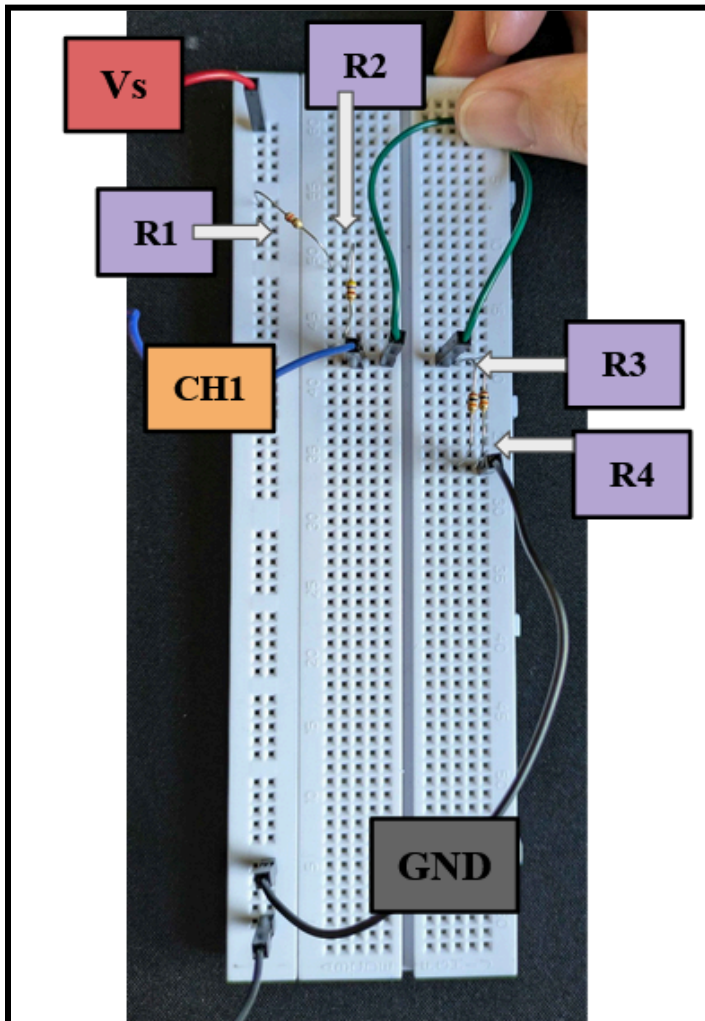
Simulation:

After simulating on LTSpice, we can see that the current at N1 is 0.000296A or 0.296mA and that the currents at R3 and R4 are both 0.000148A or 0.148mA. These simulated values appear to match the values calculated with the current divider rule.



```
--- Operating Point ---
V(n001) :      5          voltage
V(n002) :      4.34911    voltage
V(n1) :        2.95858    voltage
I(R1) :       -0.000295858 device_current
I(R2) :       -0.000295858 device_current
I(R3) :        0.000147929 device_current
I(R4) :        0.000147929 device_current
I(V1) :       -0.000295858 device_current
```

Measurement:



Made the circuit above on a breadboard like this:

Where V_s is the voltage source of 5V,

Ch1 is measuring voltage at N1,

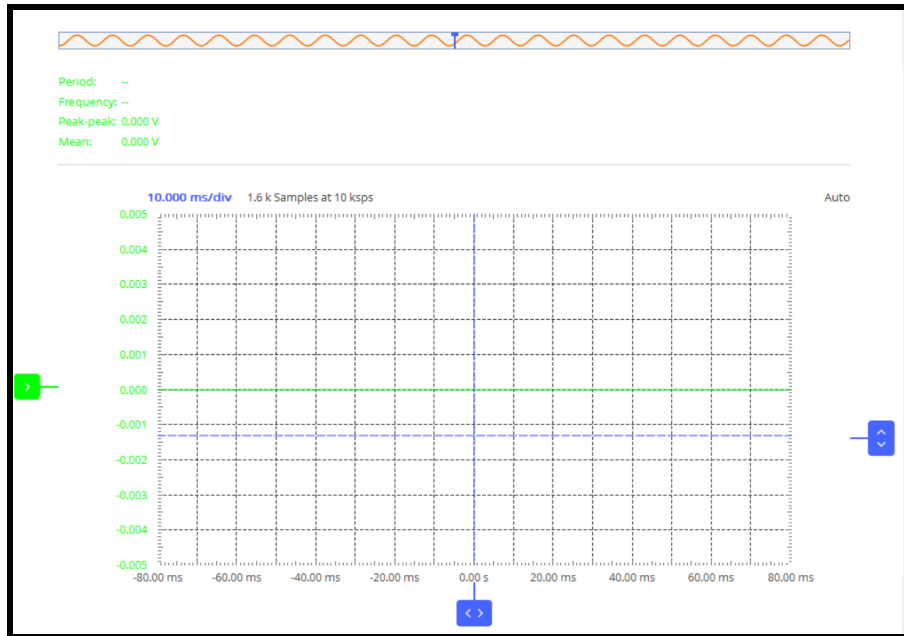
$R1 = 2.2k\Omega$

$R2 = 4.7k\Omega$

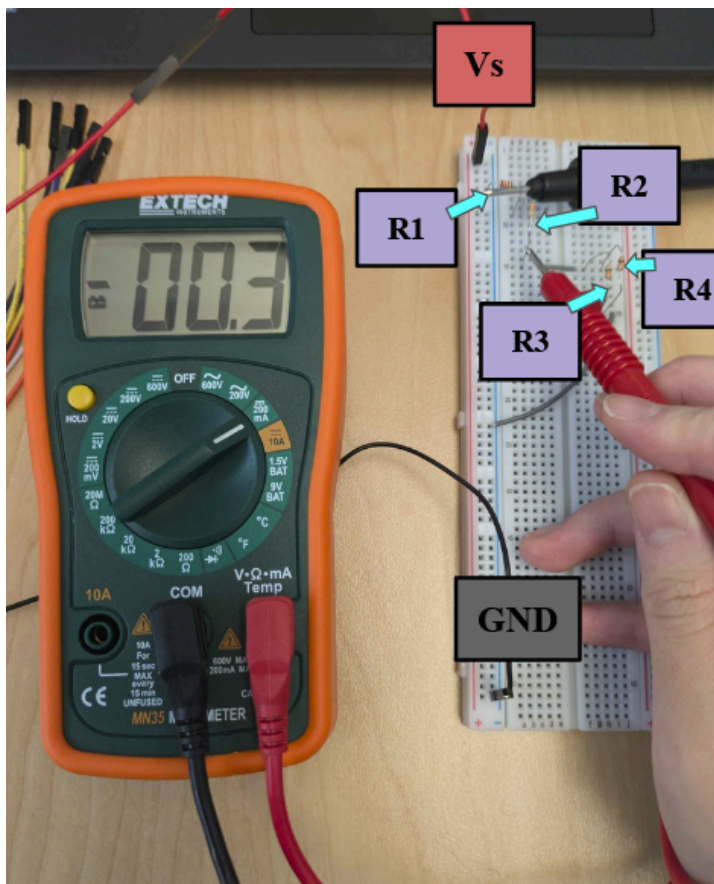
$R3 = 20k\Omega$

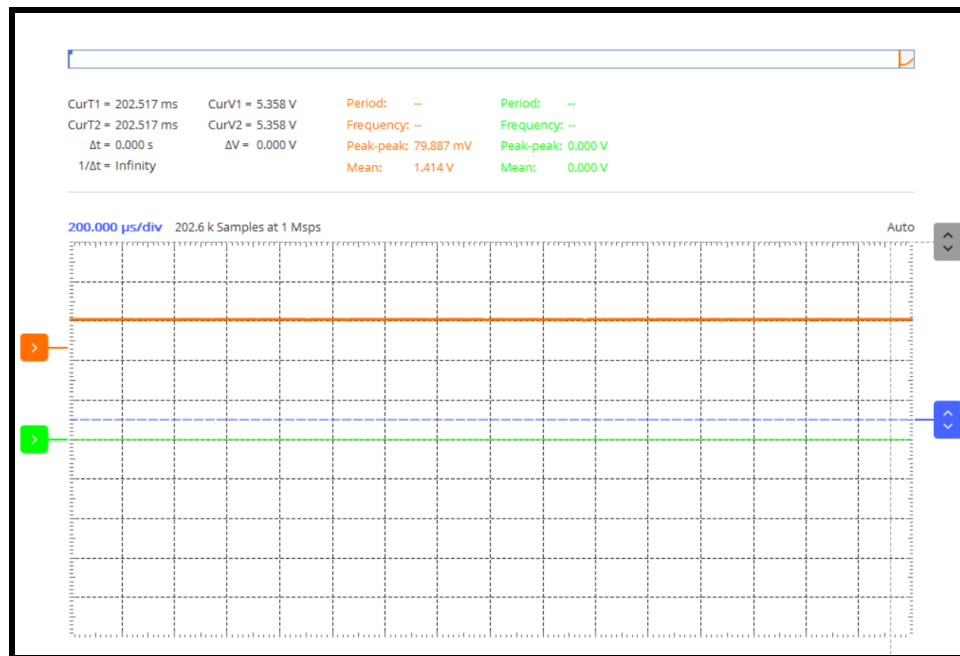
$R4 = 20k\Omega$

And GND is ground.

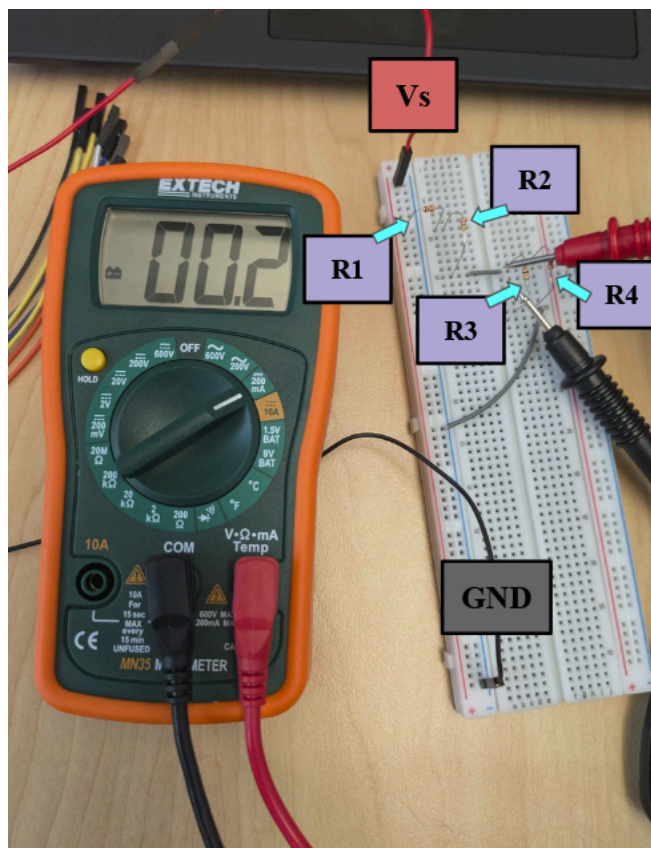


This is the measurement of current at N1 resulting in 0mA on scopy, and so I measured with a multimeter the current which was 0.3mA





The current for R3(and R4) is around 0mA on scopy and on the multimeter was 0.2mA



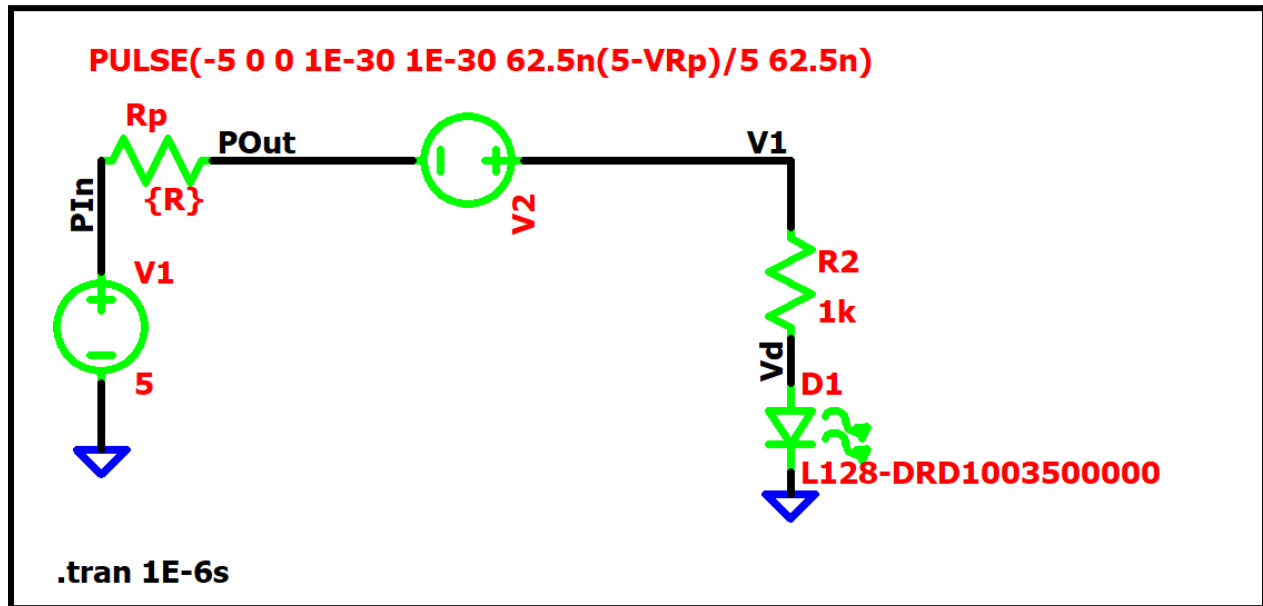
Discussion (and answer related questions in Alpha Lab):

	Analysis	Simulation	Measurement/ Experimental	Percent Error (%)
Current at N1 (mA)	0.296	0.296	0.3	1.35%
Current at R3 (mA)	0.148	0.148	0.2	35.14%
Current at R4 (mA)	0.148	0.148	0.2	35.14%

The resulting values of current measurements are somewhat as expected as the current measurement at N1 is almost the exact same as the expected value. However, the current measurement for R3 and R4 has a large percent error likely due to the multimeter rounding its value measured as well as being in units of mA results in more error than being in amps due to decimal places.

6. Pulse-Width Modulation (PWM) Control of an LED

Circuit Schematic:



Although we can't run this simulation in this state as LTSpice doesn't allow the T_{on} to be a variable, this best represents our schematic. We used the standard Arduino frequency of 16 MHz, which produces a period of 62.5 ns. The circuit will adjust its T_{on} , and therefore its duty cycle, based on the input voltage, P_{Out} , which will depend on the resistance across the potentiometer; this equation is $T_{on} = 62.5ns(V_{P_{Out}}/5) = 62.5ns((5 - V_{Rp})/5)$. The initial voltage and maximum voltage will be shifted due to the way Arduino outputs its voltage independently from the input, unlike LTSpice. The voltages will be -5 and 0 in order to best represent the shift.

Description:

We will be proving the concept of using a voltage divider containing a potentiometer to control the output of a microcontroller, which is a resistor in series with an LED. Using the potentiometer, we will prove that the brightness can be adjusted. The potentiometer in the voltage divider will control the amount of voltage being input, P_{Out} , into the Arduino as the resistance of the potentiometer, R_p , increases, the input voltage decreases, and vice versa. We will code the Arduino to use this input in a formula that controls the duty cycle of the PWM

signals. This will change the brightness of the LED by changing the average voltage the Arduino is outputting. There will also be a current-limiting resistor, R_2 , in series with the LED to ensure there is no damage done to the LED.

Analysis:

Use the formula for a voltage divider to calculate the voltage of R resulting from the potentiometer at a given resistance. We used a red LED meaning the forward voltage of the LED will be close to 1.8 V, meaning this will be the voltage across the LED. To demonstrate the current, we took measurements at 3 resistances of $5k\Omega$, $0k$, and $2.61k\Omega$.

Voltage divider formula: $V_{Rp} = V_s \frac{R_2}{R_1 + R_2}$

Where V_s is the voltage source of 5V,

R_1 is the potentiometer resistance closest to the input (0 to R_1),

And R_2 is the potentiometer resistance closest to ground (total - R_1)

To find the resulting current through the resistor and LED, use Ohm's law: $V = IR$

Where V is the Vout of the potentiometer

And R is a $1k\Omega$ resistor

When the potentiometer has resistance $5k\Omega$,

Plugging into the voltage divider formula, we get: $V_{Rp} = 5 \frac{5k}{5k + (5k - 5k)} = 5(1) = 5V$

$T_{on} = 62.5ns((5-5)/5) = 0$ ns so the duty cycle will equal

$(0/62.5ns) * 100\% = 0\%$

To find the current:

$$I = \frac{V}{R} = \frac{0}{1k} = 0 \text{ A or } 0mA$$

For when the potentiometer has a resistance of $2.61k\Omega$,

By plugging into the formula: $V_{Rp} = 5 \frac{2.61k}{2.61k + (5k - 2.61k)} = 5 \left(\frac{2.61k}{5k} \right) = 2.61V$

$T_{on} = 62.5ns \left(\frac{5 - 2.61}{5} \right) = 29.875ns$ so the duty cycle will equal

$(29.875ns / 62.5ns) * 100\% = 47.8\%$

To find the current:

$$I = \frac{V}{R} = \frac{2.39 - 1.8}{1k} = \frac{0.59}{1k} = 0.00059 A \text{ or } 0.59 mA$$

And for when the potentiometer has a resistance of $0k\Omega$,

Plugging in we get, $V_{Rp} = 5 \frac{0k}{0 + (5k - 0k)} = 5(0) = 0V$

$T_{on} = 62.5ns \left(\frac{5 - 0}{5} \right) = 62.5ns$ so the duty cycle will equal

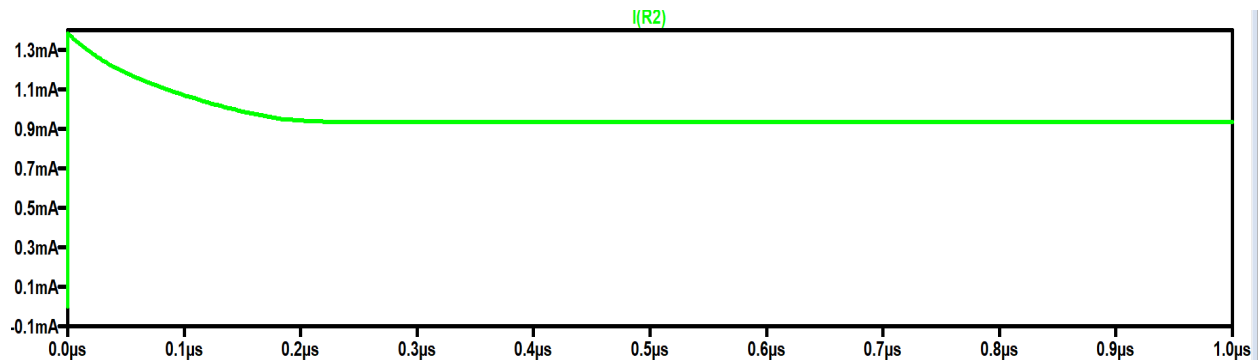
$(62.5ns / 62.5ns) * 100\% = 100\%$

To find the current:

$$I = \frac{V}{R} = \frac{5 - 1.8}{1k} = \frac{3.2}{1k} = 0.0032 A \text{ or } 3.2 mA$$

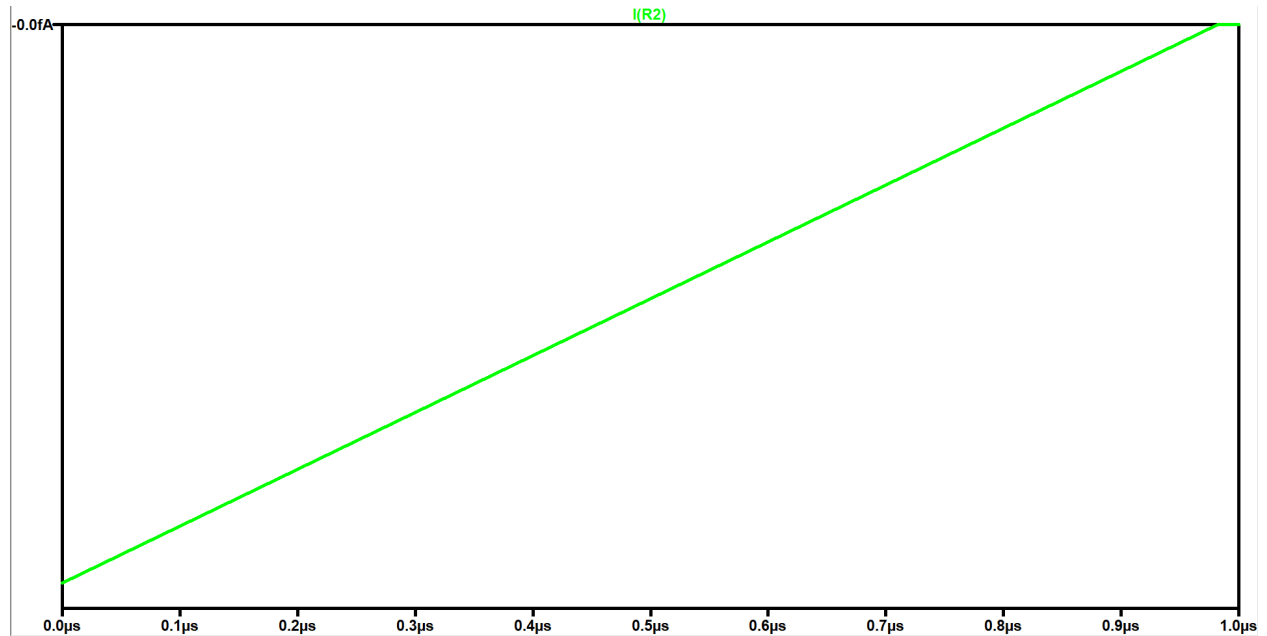
Simulation:

For when potentiometer is $2.61k\Omega$:



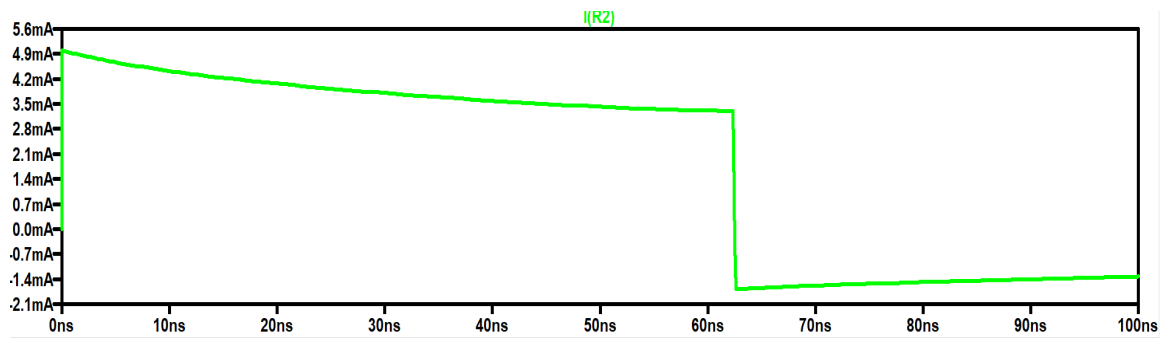
Average current of R_2 is around $0.95mA$

When potentiometer is $5k\Omega$:



Average current of R_2 is 0 mA.

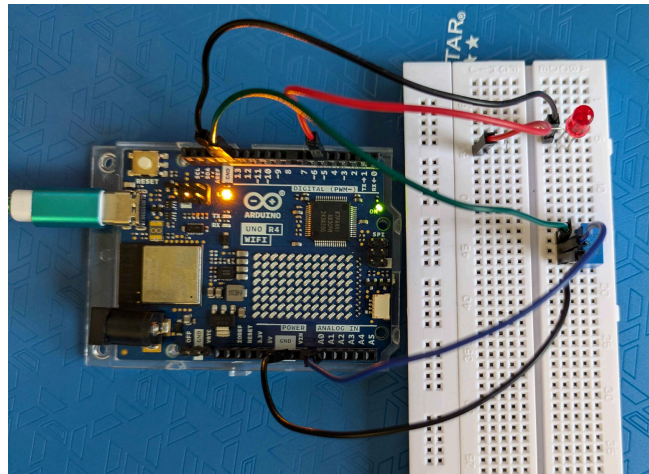
When potentiometer is $0k\Omega$:



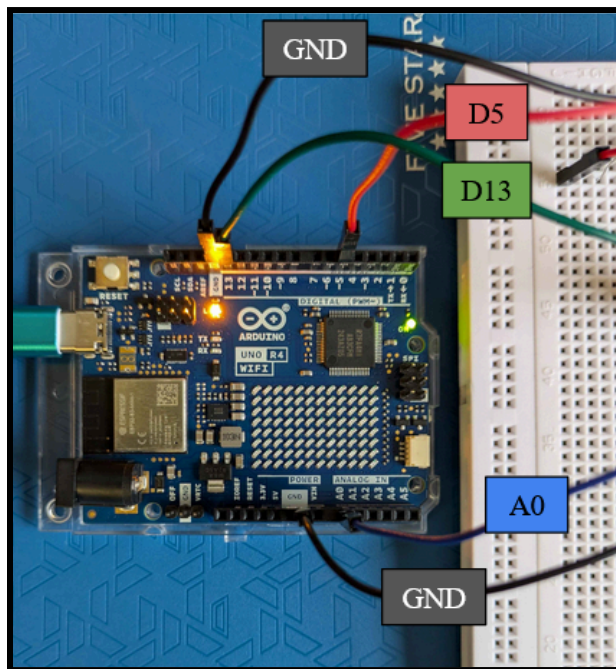
Average current of R_2 for one cycle is about 3.7 mA.

Measurement:

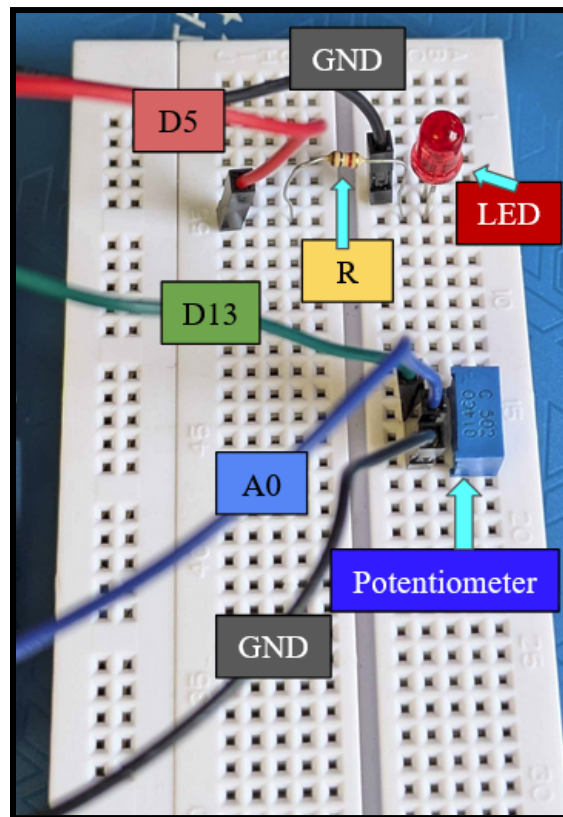
Full diagram of circuit:



D describes pins, ex: D5 is pin 5

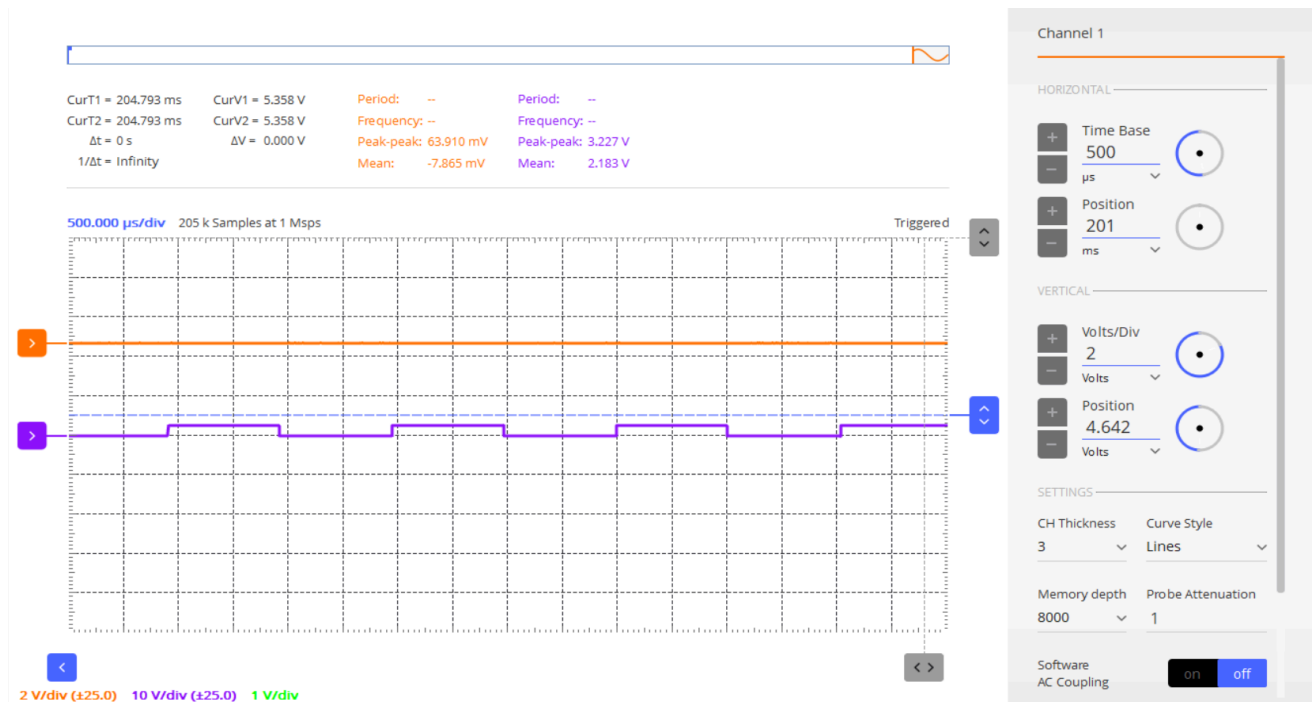
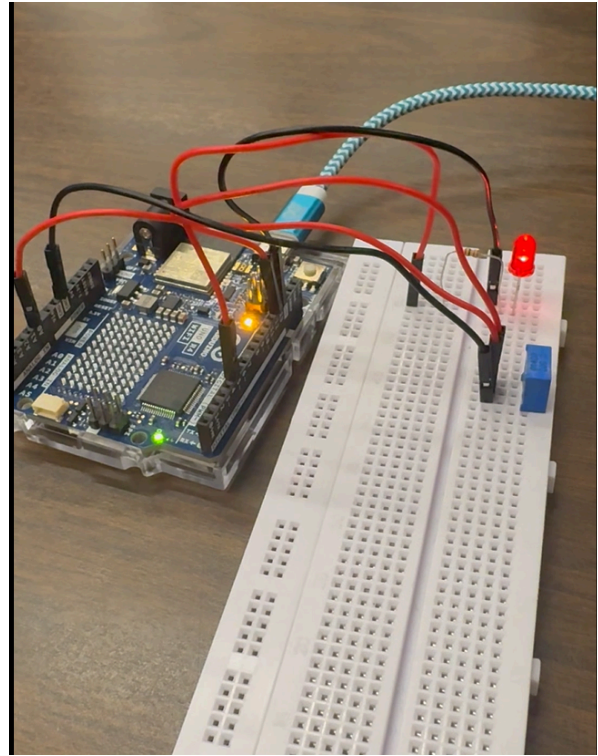
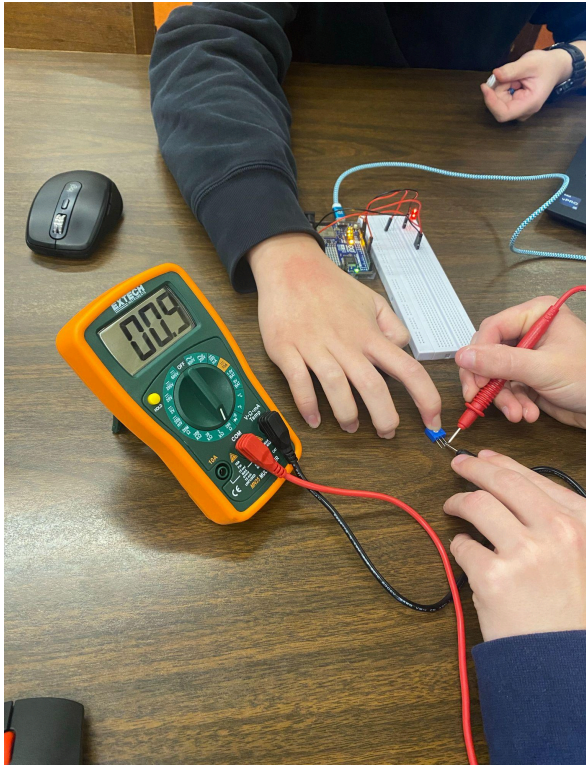


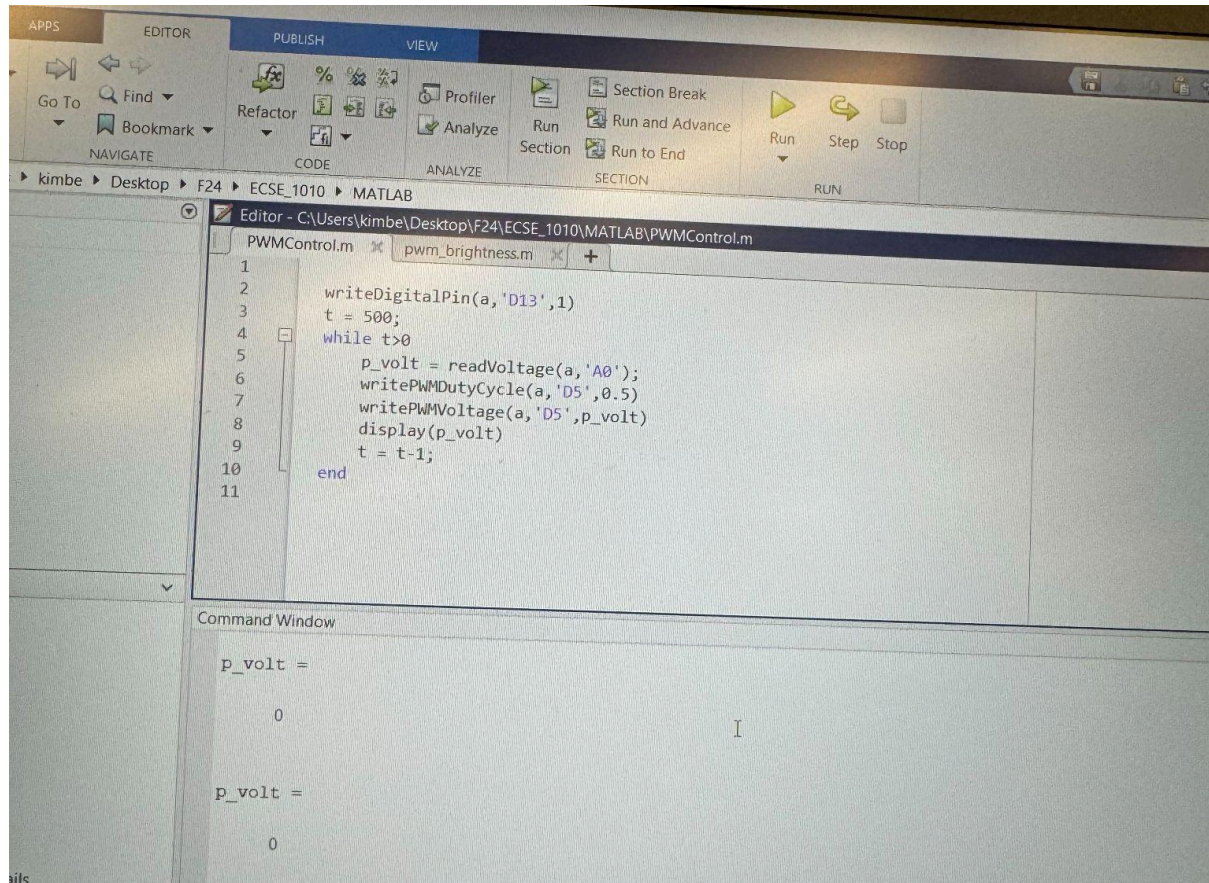
$R = 1k\Omega$



Potentiometer: max resistance of $5k\Omega$

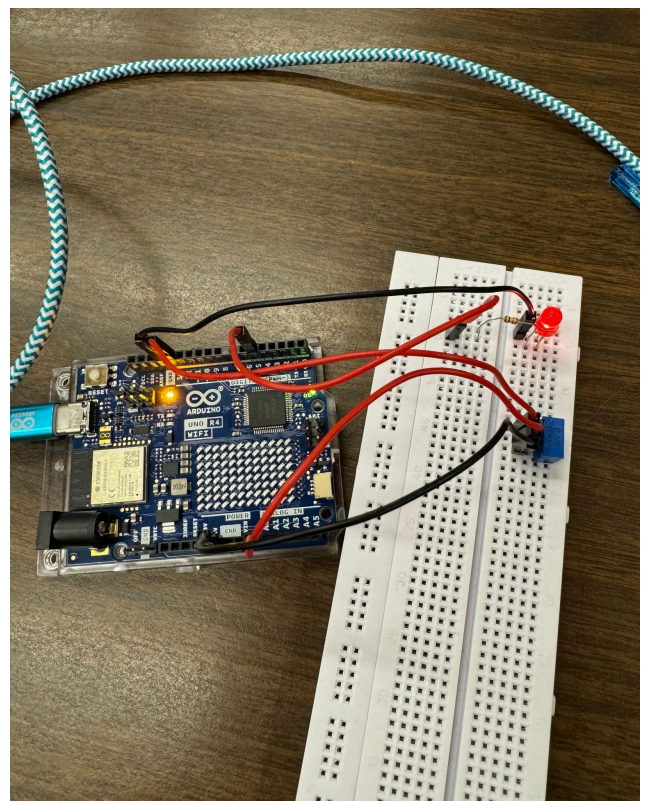
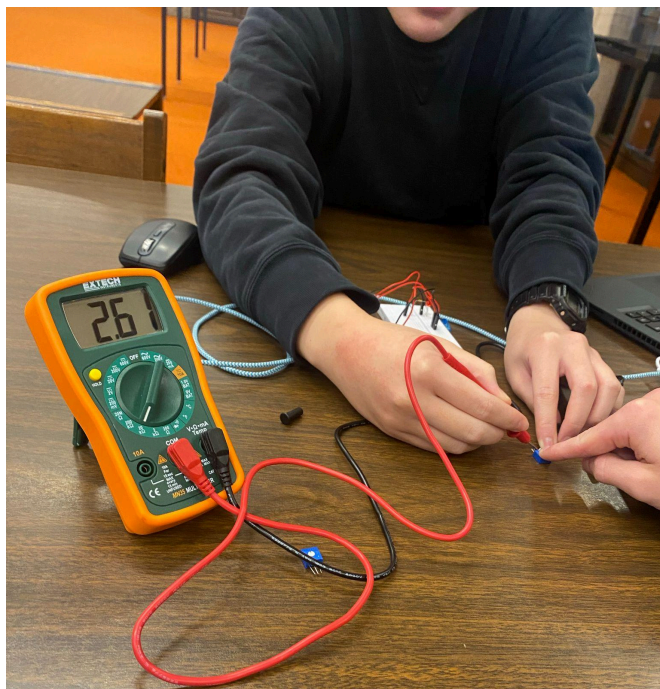
0Ω

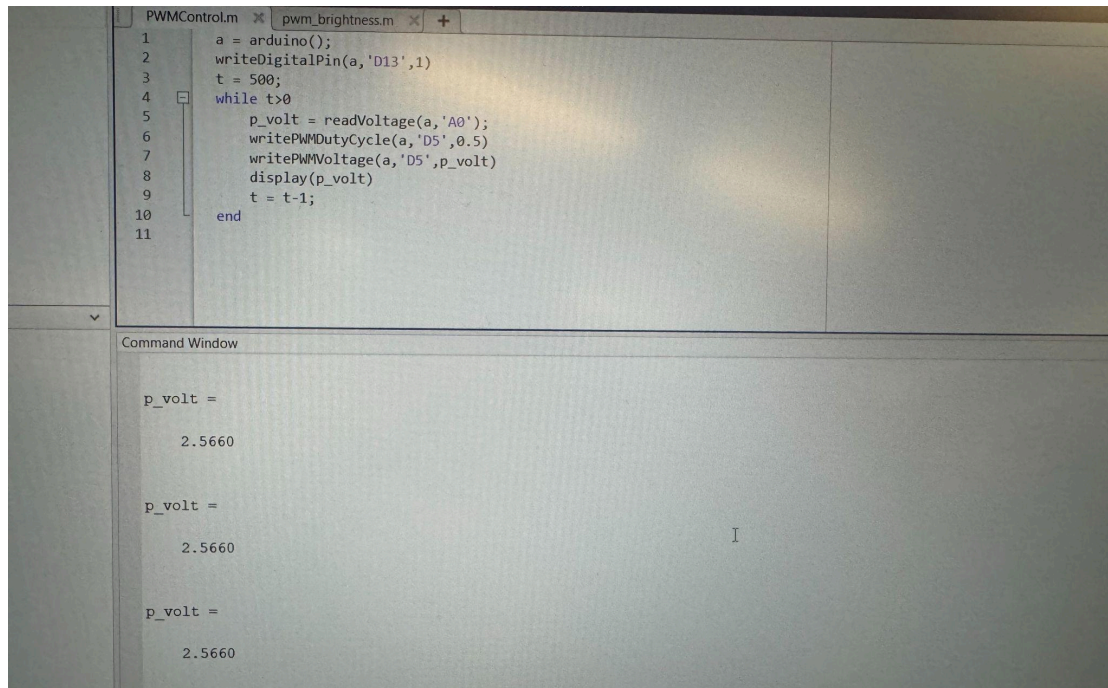




When the potentiometer has 0 resistance, the voltage is measured at -7.9 mV, which is basically 0, and the voltage across the current limiting resistor is 2.183 V, indicating the current across it and the LED is $i = 2.183 \text{ V} / 1\text{k} = 2.183 \text{ mA}$. As you can see, the LED has a constant, bright red appearance.

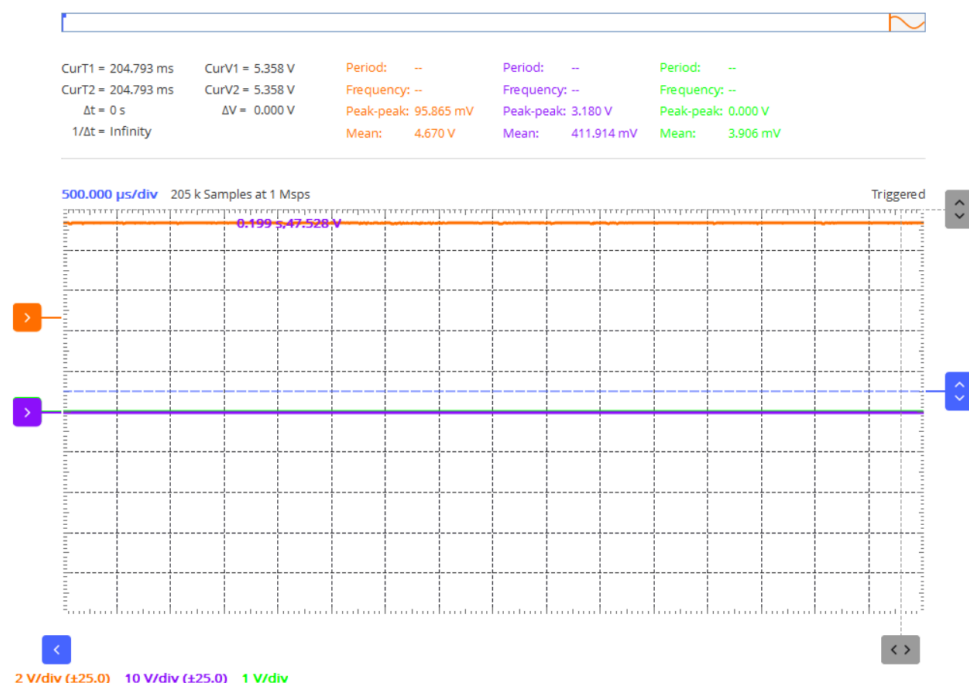
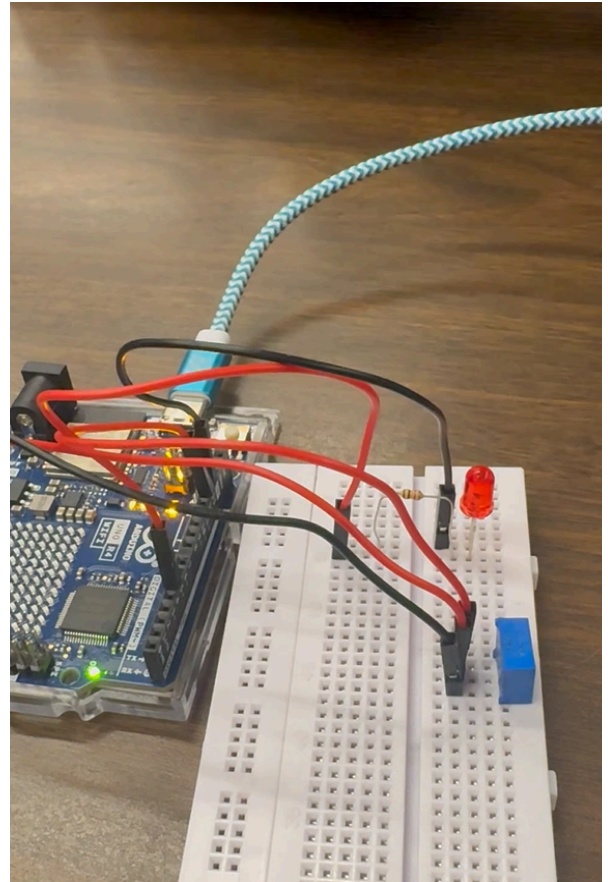
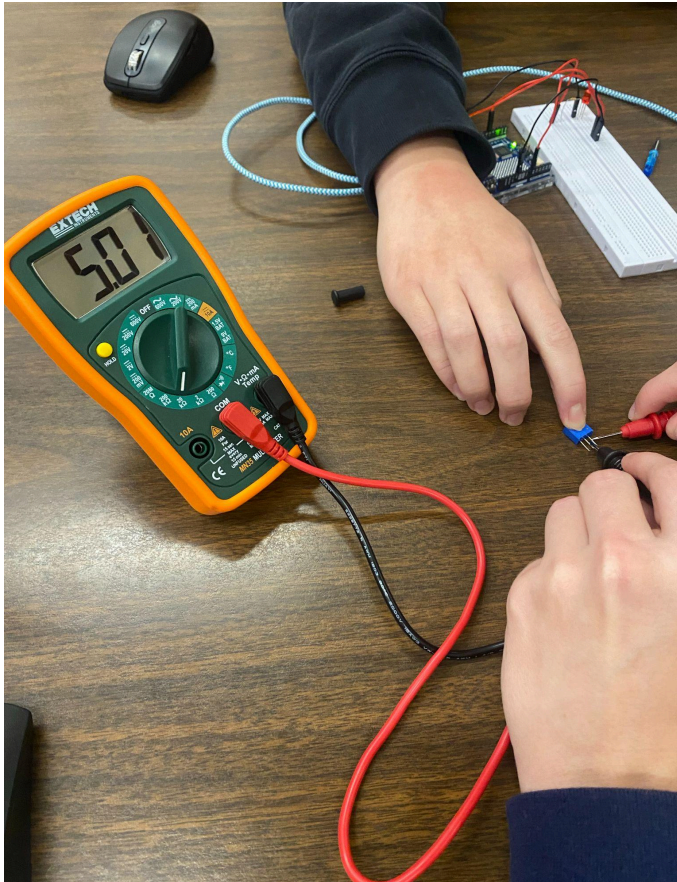
2.61k Ω

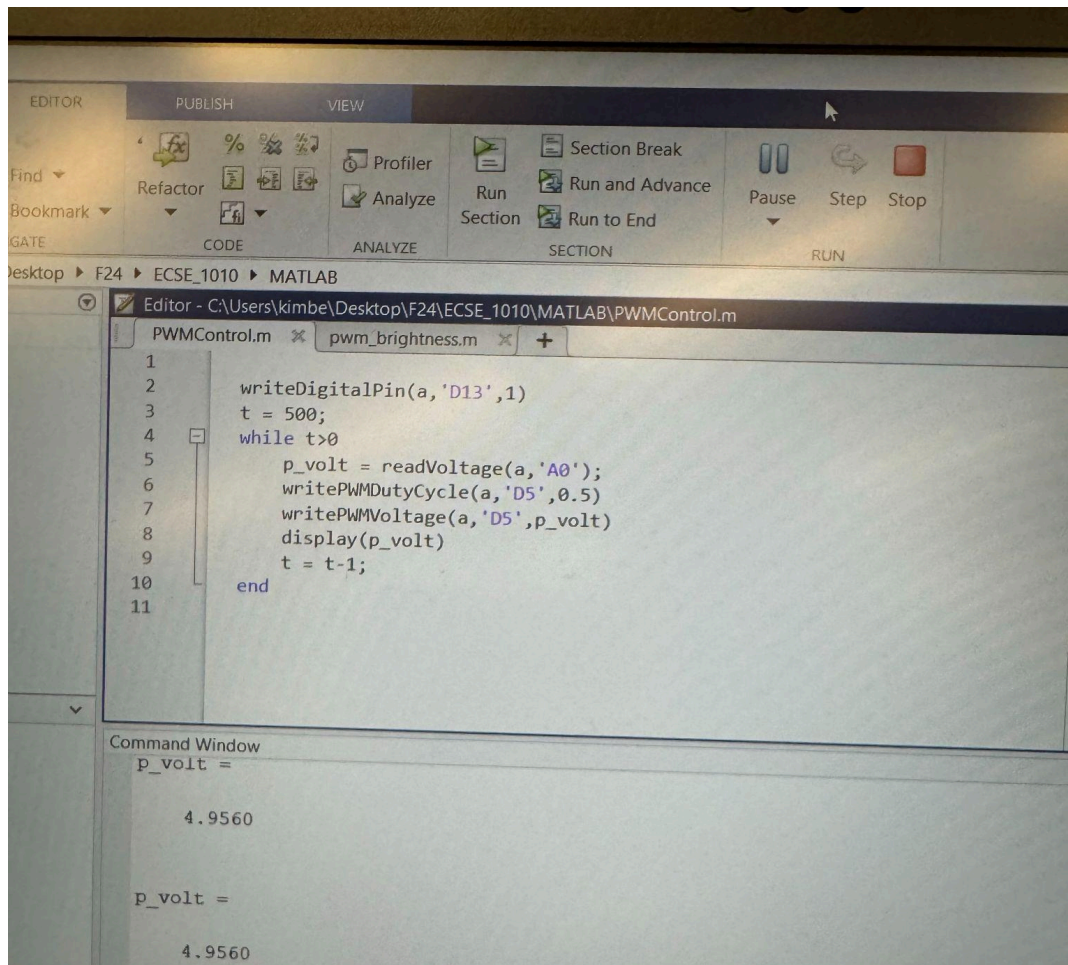




When the potentiometer has $2.61\text{k}\Omega$ resistance, the voltage is measured at 2.29 V and the voltage across the current limiting resistor is 1.348 V, indicating the current across it and the LED is $i = 1.348\text{ V} / 1\text{k} = 1.348\text{ mA}$. As you can see, the LED has a constant, red appearance with medium brightness.

5k Ω





When the potentiometer has $5k\ \Omega$ resistance, the voltage is measured at 4.67 V, and the voltage across the current limiting resistor is 412 mV, indicating the current across it and the LED is $i = 412\text{ mV} / 1k = 0.412\text{ mA}$. As you can see, the LED has low brightness.

Discussion (and answer related questions in Alpha Lab):

When $R_{pot} = 5k\Omega$

	Analysis	Simulation	Measurement	Percent Error
Voltage of potentiometer (V)	5	5	4.67	6.60%
Current through R and LED(mA)	0	0	0.412	N/A

When $R_{pot} = 0k\Omega$

	Analysis	Simulation	Measurement	Percent Error
Voltage of potentiometer (V)	0	0	-0.007	N/A
Current through R and LED(mA)	3.2	2.6	2.183	16.04%

When $R_{pot} = 2.61k\Omega$

	Analysis	Simulation	Measurement	Percent Error
Voltage of potentiometer (V)	2.61	2.61	2.29	12.26%
Current through R and LED (mA)	0.59	0.95	1.348	15.71%

From our results for each value of resistance of the potentiometer, while there is a considerable percentage of error, it isn't large enough to say that our values from analysis, simulation, and measurement aren't consistent at all. The largest percent errors are in the current values, which we chose to calculate in units of mA instead of A. Due to this unit difference, the percent error increases as the value of the measurements excusing the units is larger than they would be as just amps. Some values we simply couldn't calculate a percent error for since they were being compared to 0, which results in a calculation error as you cannot divide by 0. Another small but possible reason for why our values don't match up exactly are the values we use in simulation

and analysis are theoretical values, and specifically for our simulation on LT Spice, the values had to be rounded, and we couldn't account for the correct values of the LED.